

Highlighted Manuscript: Change and Deletion Index

Baseline: 33f5e3d37a85641b13bffa7fc34b94eb5d35c47. Selected as the latest manuscript before August 13 under the author's instruction. Yellow marks show inserted/replaced rendered words; orange frames identify pages containing changed figure assets. The clean PDF text is unchanged by annotations. Source-only layout/preamble edits are preserved in reviewed-baseline-source.diff. This index records removed text that cannot be highlighted in the revised manuscript.

Change 1: replace | old pp. [1] | new pp. [1]

Before (including deletions): SPICE

After: Structural Circuit

Change 2: replace | old pp. [1] | new pp. [1]

Before (including deletions): Automated digitization of

After: Recovering a netlist from a

Change 3: replace | old pp. [1] | new pp. [1]

Before (including deletions): circuit schematics into simulation-ready SPICE netlists remains challenging because connectivity recovery is brittle under noisy

After: schematic requires deciding which

Change 4: replace | old pp. [1] | new pp. [1]

Before (including deletions): detections.

After: strokes connect which component terminals.

Change 5: replace | old pp. [1] | new pp. [1]

Before (including deletions): that converts a scanned schematic into a SPICE netlist through component detection, an

After: for this structural connectivity task. An

Change 6: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: extractor masks components before binarization and returns

Change 7: replace | old pp. [1] | new pp. [1]

Before (including deletions): extractor,

After: segments with explicit endpoints. A typed endpoint graph links endpoints, pins and wire bodies under scale-relative tolerances,

Change 8: delete | old pp. [1] | new pp. []

Before (including deletions): graph-based join that resolves electrical connectivity. The main contribution is the joining stage: an endpoint-graph model over wire endpoints and component pins with five edge types, combined with

After: (none)

Change 9: replace | old pp. [1] | new pp. [1]

Before (including deletions): that recovers dropped connections through min- cost b-matching while bounding over-merge with per-pin budgets and self-loop guards.

After: step reconnects floating pins by bounded assignment.

Change 10: replace | old pp. [1] | new pp. [1]

Before (including deletions): also introduce a synthetic perturbation protocol and, to our knowledge, the first human-verified net-level

After: evaluate on two real-image

Change 11: replace | old pp. [1] | new pp. [1]

Before (including deletions): benchmark for hand-drawn circuits, covering

After: benchmarks from CGHD-1152:

Change 12: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: with human-verified nets, and 164 held-out photographs

Change 13: replace | old pp. [1] | new pp. [1]

Before (including deletions): CGHD-1152.

After: 24 drafters whose reference nets we derive from the dataset's own stroke and symbol annotations. With annotated component boxes, the join reaches component-pair micro-F1 0.890 on the human-verified set and 0.711 on the held-out photographs. On both benchmarks it outperforms every deterministic baseline, including connected-component tracing and Hough line fitting, in paired tests with Holm correction. With annotated wires, its score stays within 0.005 of the native value when the images are rescaled from 0.5× to 3×. Replacing the annotated boxes with a trained component detector gives micro-F1 0.627, and most of that loss comes from missed components.

Change 14: replace | old pp. [1] | new pp. [1]

Before (including deletions): occlusion-first

After: wire

Change 15: delete | old pp. [1] | new pp. []

Before (including deletions): wire-detection

After: (none)

Change 16: delete | old pp. [1] | new pp. []

Before (including deletions): =

After: (none)

Change 17: replace | old pp. [1] | new pp. [1]

Before (including deletions): scanned images. On

After: images; on

Change 18: replace | old pp. [1] | new pp. [1]

Before (including deletions): human-verified net-level benchmark,

After: held-out photographs, wire extraction is

Change 19: replace | old pp. [1] | new pp. [1]

Before (including deletions): best join reaches connectivity micro-F1 = 0.890, outperforming the prior completion default (0.83), a legacy radius union-find join (0.67), a connected-component net-tracing baseline (0.62), and a Hough- plus-proximity line tracer (0.81), all under identical detected wires. Running the join on perfect wire labels does not improve micro-F1, showing that connectivity, not wire detection, remains the bottleneck.

After: main source of error.

Change 20: delete | old pp. [1] | new pp. []

Before (including deletions): first- party comparison shows a strong

After: (none)

Change 21: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: given the same component boxes

Change 22: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: micro-F1

Change 23: replace | old pp. [1] | new pp. [1]

Before (including deletions): same images, statistically indistinguishable

After: human-verified set: the pooled difference

Change 24: replace | old pp. [1] | new pp. [1]

Before (including deletions): but far more expensive

After: is not significant,

Change 25: replace | old pp. [1] | new pp. [1]

Before (including deletions): without structural guarantees, supporting explicit geometric connectivity recovery for deployable circuit digitization.

After: per-image scores favour the model. We release the code, both benchmarks and the evaluation scripts.

Change 26: replace | old pp. [1] | new pp. [1]

Before (including deletions): SPICE,

After: structural netlist,

Change 27: replace | old pp. [1] | new pp. [1]

Before (including deletions): Analog

After: A

Change 28: replace | old pp. [1] | new pp. [1]

Before (including deletions): schematics encode functional information (component

After: schematic encodes component

Change 29: replace | old pp. [1] | new pp. [1]

Before (including deletions): values,

After: component values

Change 30: replace | old pp. [1] | new pp. [1]

Before (including deletions): topological connections)

After: the connections between components

Change 31: replace | old pp. [1] | new pp. [1]

Before (including deletions): visual format designed

After: drawing made

Change 32: replace | old pp. [1] | new pp. [1]

Before (including deletions): human interpretation. Converting scanned

After: people. A netlist records the same connections in a form that software can check, simulate and search. Con-verting hand-drawn

Change 33: delete | old pp. [1] | new pp. []

Before (including deletions): machine-readable SPICE (Simula- tion Program with Integrated Circuit Emphasis)

After: (none)

Change 34: replace | old pp. [1] | new pp. [1]

Before (including deletions): en- ables automated simulation, design verification,

After: would let course material, lab notebooks

Change 35: replace | old pp. [1] | new pp. [1]

Before (including deletions): large- scale circuit analysis. While recent advances

After: legacy designs enter these tools without manual re-entry. This paper addresses the structural part of that conversion: given the components

Change 36: replace | old pp. [1] | new pp. [1]

Before (including deletions): object detec- tion have made component recognition increasingly reliable, the joining problem, determining which wire segments con- nect to

After: an image, decide

Change 37: replace | old pp. [1] | new pp. [1]

Before (including deletions): pins and to each other, remains the primary failure mode in end-to-end digitization pipelines. The challenge

After: terminals share an electrical node. We call this step joining. Joining

Change 38: replace | old pp. [1] | new pp. [1]

Before (including deletions): threefold.

After: hard for three reasons.

Change 39: delete | old pp. [1] | new pp. []

Before (including deletions): inherently

After: (none)

Change 40: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: across drafters,

Change 41: delete | old pp. [1] | new pp. []

Before (including deletions): drawn by different drafters

After: (none)

Change 42: replace | old pp. [1] | new pp. [1]

Before (including deletions): components occupy bounding boxes

After: component symbols overlap the wire endpoints

Change 43: replace | old pp. [1] | new pp. [1]

Before (including deletions): overlap with wire endpoints, creating ambiguity about whether

After: attach to them, so

Change 44: replace | old pp. [1] | new pp. [1]

Before (including deletions): connects

After: near a component may belong

Change 45: replace | old pp. [1] | new pp. [1]

Before (including deletions): a component's pin

After: one of its pins

Change 46: replace | old pp. [1] | new pp. [1]

Before (including deletions): passes nearby.

After: pass by.

Change 47: replace | old pp. [1] | new pp. [1]

Before (including deletions): fragments,

After: fragments

Change 48: replace | old pp. [1] | new pp. [1]

Before (including deletions): crossing wires create complex connectivity patterns that simple radius-based approaches cannot resolve without over-merging.

After: wire crossings all look alike to a proximity rule. Most image-to-netlist systems erase the components and

Change 49: replace | old pp. [2] | new pp. [2]

Before (including deletions): 88.5%

After: 89.0%

Change 50: replace | old pp. [2] | new pp. [2]

Before (including deletions): SPICE

After: Structural

Change 51: replace | old pp. [2] | new pp. [2]

Before (including deletions): Pin discovery + simulation

After: Inferred pin-to-node map

Change 52: replace | old pp. [2] | new pp. [2]

Before (including deletions): End-to-end pipeline

After: Pipeline

Change 53: replace | old pp. [2] | new pp. [2]

Before (including deletions): scanning circuit diagram digitization. Our contributions span the

After: structural circuit-netlist recovery. The highlighted stages extract

Change 54: replace | old pp. [2] | new pp. [2]

Before (including deletions): extraction

After: endpoints

Change 55: delete | old pp. [2] | new pp. []

Before (including deletions): joining stages; the

After: (none)

Change 56: replace | old pp. [2] | new pp. [2]

Before (including deletions): (dashed red) determines which extracted wire segments connect to which component pins and is the dominant failure mode in end-to-end digitization.

After: them into nets.

Change 57: delete | old pp. [2] | new pp. []

Before (including deletions): real

After: (none)

Change 58: replace | old pp. [2] | new pp. [2]

Before (including deletions): circuit images,

After: circuits,

Change 59: replace | old pp. [2] | new pp. [2]

Before (including deletions): our

After: the

Change 60: replace | old pp. [2] | new pp. [2]

Before (including deletions): (connectivity

After: (component-pair

Change 61: replace | old pp. [2] | new pp. [2]

Before (including deletions): (36

After: (35

Change 62: replace | old pp. [2] | new pp. [2]

Before (including deletions): 12 nets); right:

After: 7 nets). Right:

Change 63: replace | old pp. [2] | new pp. [2]

Before (including deletions): (12

After: (11

Change 64: replace | old pp. [2] | new pp. [2]

Before (including deletions): shows, in order: original scan,

After: shows the input,

Change 65: replace | old pp. [2] | new pp. [2]

Before (including deletions): close,

After: closing,

Change 66: replace | old pp. [2] | new pp. [2]

Before (including deletions): (red overlay),

After: (red)

Change 67: replace | old pp. [2] | new pp. [2]

Before (including deletions): every recovered

After: each

Change 68: replace | old pp. [2] | new pp. [2]

Before (including deletions): is a distinct colour drawn along

After: has

Change 69: replace | old pp. [2] | new pp. [2]

Before (including deletions): conductors; component bounding

After: own colour;

Change 70: replace | old pp. [2] | new pp. [2]

Before (including deletions): ground-truth labels (as used throughout the net-level evaluation, isolating the join),

After: annotated components,

Change 71: replace | old pp. [2] | new pp. [2]

Before (including deletions): mark junctions,

After: are junctions

Change 72: replace | old pp. [2] | new pp. [2]

Before (including deletions): mark

After: are

Change 73: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: read nets from the connected components or contours of the remaining wire pixels [3], [5].

Change 74: replace | old pp. [2] | new pp. [2]

Before (including deletions): natural alternative is

After: single stroke gap then splits a net, and a single touching stroke merges two. Other systems pass the image

Change 75: delete | old pp. [2] | new pp. []

Before (including deletions): prompt

After: (none)

Change 76: replace | old pp. [2] | new pp. [2]

Before (including deletions): vision-language

After: language

Change 77: replace | old pp. [2] | new pp. [2]

Before (including deletions): (VLM) to

After: and

Change 78: replace | old pp. [2] | new pp. [2]

Before (including deletions): the schematic

After: connectivity from its answer [4]. We keep an explicit representation of every wire endpoint

Change 79: replace | old pp. [2] | new pp. [2]

Before (including deletions): emit its netlist directly. We test this in Section IV-F: a strong VLM attains connectivity accuracy statistically indistinguishable from our

After: decide each connection with

Change 80: replace | old pp. [2] | new pp. [2]

Before (including deletions): pipeline, but it does so at orders-of-magnitude higher cost, returns free-form text rather than a structured netlist, offers no structural guarantee against electrically invalid output (e.g. shorted two-terminal parts), and drops connections on the most complex circuits. These properties motivate an explicit, inspectable geometric pipeline whose output is a netlist by construction and whose accuracy matches the VLM's while being deterministic and simulatable. We address these challenges with a complete deterministic pipeline whose novel elements are: (1) an

After: rules that can be inspected. This paper makes four contributions: 1) An

Change 81: replace | old pp. [2] | new pp. [2]

Before (including deletions): masks detected components

After: fills each component region with its local median colour

Change 82: replace | old pp. [2] | new pp. [2]

Before (including deletions): binarization

After: Sauvola binarization

Change 83: replace | old pp. [2] | new pp. [2]

Before (including deletions): estimates per-segment endpoints, producing a line-segment-with-endpoints representation (rather than tracing nets directly into connected components, as prior systems do) that is what makes graph-based joining possible; (2) an

After: returns line segments with PCA-estimated endpoints. It reaches wire-detection F1 0.976 on 134 CGHD-1152 images. 2) A typed

Change 84: delete | old pp. [2] | new pp. []

Before (including deletions): model that constructs a graph over both wire endpoints and component pins,

After: (none)

Change 85: replace | old pp. [2] | new pp. [2]

Before (including deletions): types capturing distinct

After: rules, component-first pin assignment and scale-relative tolerances, followed by degree-budget completion, which reconnects floating pins through a bounded-capacity assignment with a shared-component guard. 3) Two real-image

Change 86: replace | old pp. [2] | new pp. [2]

Before (including deletions): patterns; and (3) a degree-budget completion algorithm that identifies and reconnects dropped connections using constrained optimization. We evaluate on the CGHD-1152 dataset [1]

After: benchmarks: 31 CGHD-1152 images

Change 87: replace | old pp. [2] | new pp. [2]

Before (including deletions): both a synthetic error injection framework that enables systematic robustness analysis

After: human-verified nets,

Change 88: replace | old pp. [2] | new pp. [2]

Before (including deletions): human-verified net-level benchmark.

After: 164-image held-out benchmark covering 24 drafters whose reference nets are derived from the CGHD authors' own stroke-segmentation maps and symbol polygons. The derived reference agrees with the human nets on 17 overlapping images at micro-F1 0.939. 4) An evaluation protocol with paired significance tests under Holm correction, complexity and component-size strata, controlled image rescaling, end-to-end runs with a trained component detector, and a vision-language model (VLM) reference that receives the same component boxes as our join. The output is a structural netlist; recognizing component values and device models is outside the scope of this work. Section II reviews related work, Section III describes the method and Section IV the experimental setup. Section V reports results, and Section VI discusses them together with the scope and limitations of the study.

Change 89: replace | old pp. [2] | new pp. [2]

Before (including deletions): The end-to-end conversion of

After: Several systems convert

Change 90: replace | old pp. [2] | new pp. [2]

Before (including deletions): machine-readable netlists has progressed rapidly.

After: machine- readable netlists.

Change 91: replace | old pp. [2] | new pp. [2]

Before (including deletions): pattern-recognition

After: pattern- recognition

Change 92: replace | old pp. [2] | new pp. [2]

Before (including deletions): digi- tizing circuit

After: printed

Change 93: replace | old pp. [2] | new pp. [2]

Before (including deletions): com- bining

After: combining

Change 94: replace | old pp. [2] | new pp. [2]

Before (including deletions): [3], the clos- est contemporary system,

After: [3]

Change 95: replace | old pp. [2] | new pp. [2]

Before (including deletions): connected-component

After: connected- component

Change 96: replace | old pp. [2] | new pp. [2]

Before (including deletions): invokes

After: uses

Change 97: replace | old pp. [2] | new pp. [2]

Before (including deletions): vision-language model (VLM)

After: VLM only

Change 98: delete | old pp. [2] | new pp. []

Before (including deletions): confines the VLM to symbolic labeling and

After: (none)

Change 99: replace | old pp. [2] | new pp. [2]

Before (including deletions): proximity

After: proxim- ity

Change 100: replace | old pp. [2] | new pp. [2]

Before (including deletions): trusting

After: from

Change 101: replace | old pp. [2] | new pp. [2]

Before (including deletions): VLM with connectivity, an insight we share (Section II).

After: VLM.

Change 102: delete | old pp. [2] | new pp. []

Before (including deletions): (Auto-SPICE)

After: (none)

Change 103: replace | old pp. [2] | new pp. [2]

Before (including deletions): produce

After: generate

Change 104: replace | old pp. [2] | new pp. [2]

Before (including deletions): netlists from

After: descriptions of

Change 105: replace | old pp. [2] | new pp. [2]

Before (including deletions): Most directly comparable to our setting, the

After: The

Change 106: delete | old pp. [2] | new pp. []

Before (including deletions): (IEEE Access, 2026)

After: (none)

Change 107: replace | old pp. [2] | new pp. [2]

Before (including deletions): a fully

After: an

Change 108: replace | old pp. [2] | new pp. [2]

Before (including deletions): image-to- SPICE

After: image-to-SPICE

Change 109: replace | old pp. [2] | new pp. [2]

Before (including deletions): circuits, pairing

After: circuits. It pairs

Change 110: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: 2 VOLUME 11, 2023

Change 111: replace | old pp. [2] | new pp. [3]

Before (including deletions): module (components

After: module, in which components

Change 112: replace | old pp. [2] | new pp. [3]

Before (including deletions): pixels) and, as we do,

After: pixels, and it

Change 113: delete | old pp. [2] | new pp. []

Before (including deletions): automated

After: (none)

Change 114: replace | old pp. [2, 3] | new pp. [3]

Before (including deletions): simula- 2 VOLUME 11, 2023 tion

After: simulation

Change 115: replace | old pp. [3] | new pp. [3]

Before (including deletions): their

After: an

Change 116: replace | old pp. [3] | new pp. [3]

Before (including deletions): they report

After: it reports

Change 117: replace | old pp. [3] | new pp. [3]

Before (including deletions): Their

After: Its

Change 118: replace | old pp. [3] | new pp. [3]

Before (including deletions): is an instance of

After: belongs to

Change 119: replace | old pp. [3] | new pp. [3]

Before (including deletions): connected-component/contour recipe

After: connected-component family that

Change 120: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: on identical wire inputs

Change 121: replace | old pp. [3] | new pp. [3]

Before (including deletions): IV-D); because their

After: V-C); its

Change 122: delete | old pp. [3] | new pp. []

Before (including deletions): (transistor-level analog, authored in-house)

After: (none)

Change 123: delete | old pp. [3] | new pp. []

Before (including deletions): (LT- spice node-voltage pass/fail)

After: (none)

Change 124: replace | old pp. [3] | new pp. [3]

Before (including deletions): ours (discrete com- ponents on CGHD-1152, component-pair F1),

After: ours, so

Change 125: replace | old pp. [3] | new pp. [3]

Before (including deletions): numbers

After: num- bers

Change 126: replace | old pp. [3] | new pp. [3]

Before (including deletions): comparable, but the comparison shows contour/CCL node detection remains the prevailing approach, which our endpoint-graph join substantially outper- forms on a controlled, identical-input benchmark. Instance- segmentation approaches extract circuit graphs directly from pixels:

After: comparable.

Change 127: replace | old pp. [3] | new pp. [3]

Before (including deletions): [6]

After: [8]

Change 128: replace | old pp. [3] | new pp. [3]

Before (including deletions): over

After: on

Change 129: replace | old pp. [3] | new pp. [3]

Before (including deletions): in- puts,

After: inputs,

Change 130: replace | old pp. [3] | new pp. [3]

Before (including deletions): [7]

After: [9]

Change 131: replace | old pp. [3] | new pp. [3]

Before (including deletions): recent JUHCCR- v1

After: JUHCCR-v1

Change 132: replace | old pp. [3] | new pp. [3]

Before (including deletions): [8]

After: [10]

Change 133: replace | old pp. [3] | new pp. [3]

Before (including deletions): recognition

After: recog- nition

Change 134: delete | old pp. [3] | new pp. []

Before (including deletions): object

After: (none)

Change 135: replace | old pp. [3] | new pp. [3]

Before (including deletions): [9], [10]

After: [11], [12]

Change 136: replace | old pp. [3] | new pp. [3]

Before (including deletions): [11], with the ultimate goal of producing simulator-ready descriptions in

After: [13], and target simulator input

Change 137: replace | old pp. [3] | new pp. [3]

Before (including deletions): [12]. These works establish detection

After: [14]. SPICE-level modeling also supports hardware work in neighboring domains, including memristor-based neuromorphic circuits that realize brain- inspired learning [6]

Change 138: replace | old pp. [3] | new pp. [3]

Before (including deletions): recognition as largely solved;

After: memristive decision-making circuits for autonomous navigation [7]. We focus on

Change 139: replace | old pp. [3] | new pp. [3]

Before (including deletions): bottleneck shifts to reconstructing electrical connectivity, the focus of our contribution.

After: joining stage and evaluate it with both annotated and detected components.

Change 140: delete | old pp. [3] | new pp. []

Before (including deletions): hand- drawing

After: (none)

Change 141: replace | old pp. [3] | new pp. [3]

Before (including deletions): an- notations

After: annota- tions

Change 142: replace | old pp. [3] | new pp. [3]

Before (including deletions): analog/mixed- signal

After: analog/mixed-signal

Change 143: replace | old pp. [3] | new pp. [3]

Before (including deletions): [13]

After: [15]

Change 144: replace | old pp. [3] | new pp. [3]

Before (including deletions): detec- tion, while

After: detection, and

Change 145: replace | old pp. [3] | new pp. [3]

Before (including deletions): [14] is a benchmark for evaluat- ing

After: [16] evaluates

Change 146: replace | old pp. [3] | new pp. [3]

Before (including deletions): LLM capabilities

After: LLMs

Change 147: delete | old pp. [3] | new pp. []

Before (including deletions): Beyond circuits,

After: (none)

Change 148: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15]

After: [17]

Change 149: replace | old pp. [3] | new pp. [3]

Before (including deletions): an end-to-end

After: a

Change 150: replace | old pp. [3] | new pp. [3]

Before (including deletions): di- agrams,

After: diagrams,

Change 151: replace | old pp. [3] | new pp. [3]

Before (including deletions): an explicit

After: a

Change 152: replace | old pp. [3] | new pp. [3]

Before (including deletions): We use these benchmarks both as evidence of where current systems succeed (component recognition) and where they fail (connectivity), and contribute a synthetic robustness protocol that isolates the join step under controlled detector noise.

After: To our knowledge, no prior benchmark reports net-level connectivity on hand- drawn circuit images.

Change 153: replace | old pp. [3] | new pp. [3]

Before (including deletions): [16],

After: [18]

Change 154: replace | old pp. [3] | new pp. [3]

Before (including deletions): understand- ing in particular.

After: understanding.

Change 155: replace | old pp. [3] | new pp. [3]

Before (including deletions): [17]

After: [19]

Change 156: replace | old pp. [3] | new pp. [3]

Before (including deletions): using

After: with

Change 157: delete | old pp. [3] | new pp. []

Before (including deletions): (VQA)

After: (none)

Change 158: replace | old pp. [3] | new pp. [3]

Before (including deletions): connection- matrix

After: connection-matrix

Change 159: delete | old pp. [3] | new pp. []

Before (including deletions): However, multiple independent bench- marks show that general VLMs handle component recogni- tion far better than connectivity.

After: (none)

Change 160: replace | old pp. [3] | new pp. [3]

Before (including deletions): con- nection

After: connection

Change 161: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15], leading

After: [17],

Change 162: delete | old pp. [3] | new pp. []

Before (including deletions): very

After: (none)

Change 163: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 164: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 165: delete | old pp. [3] | new pp. []

Before (including deletions): and

After: (none)

Change 166: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 167: replace | old pp. [3] | new pp. [3]

Before (including deletions): [14]

After: [16]

Change 168: delete | old pp. [3] | new pp. []

Before (including deletions): MLLMs on

After: (none)

Change 169: replace | old pp. [3] | new pp. [3]

Before (including deletions): This corroborates the design choice, shared

After: These results vary

Change 170: replace | old pp. [3] | new pp. [3]

Before (including deletions): SINA [3], to keep wire-to-component

After: task and input, which motivates mea- suring VLM

Change 171: replace | old pp. [3] | new pp. [3]

Before (including deletions): out of the VLM and instead solve it with an explicit geometric model. We make this separation concrete and quantify it with

After: directly on

Change 172: replace | old pp. [3] | new pp. [3]

Before (including deletions): own Claude-VLM connectivity experiment.

After: benchmark (Section V-I).

Change 173: replace | old pp. [3] | new pp. [3]

Before (including deletions): morphological operations,

After: morphological operations

Change 174: replace | old pp. [3] | new pp. [3]

Before (including deletions): [18], typically preceded by

After: [20], usually after

Change 175: replace | old pp. [3] | new pp. [3]

Before (including deletions): thresholding [19]

After: thresholding [21]

Change 176: replace | old pp. [3] | new pp. [3]

Before (including deletions): [20].

After: [22].

Change 177: replace | old pp. [3] | new pp. [3]

Before (including deletions): dominant

After: legacy

Change 178: replace | old pp. [3] | new pp. [3]

Before (including deletions): existing pipelines

After: our pipeline

Change 179: replace | old pp. [3] | new pp. [3]

Before (including deletions): grabs all

After: takes every

Change 180: replace | old pp. [3] | new pp. [3]

Before (including deletions): pins

After: pin

Change 181: replace | old pp. [3] | new pp. [3]

Before (including deletions): (typically 30px), connected via transitive union-find [21]. This over-merges when multiple

After: (30 px), and connections are merged transitively with union-find [23]. When several

Change 182: replace | old pp. [3] | new pp. [3]

Before (including deletions): shorting

After: this rule assigns both pins of a

Change 183: replace | old pp. [3] | new pp. [3]

Before (including deletions): parts (R, C, L, D) onto

After: part to

Change 184: delete | old pp. [3] | new pp. []

Before (including deletions): SINA [3] likewise derives nets from CCL proximity, which is similarly vulnerable to over-merge under detector noise.

After: (none)

Change 185: delete | old pp. [3] | new pp. []

Before (including deletions): instead

After: (none)

Change 186: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: instead

Change 187: replace | old pp. [3] | new pp. [3]

Before (including deletions): incorporates

After: uses

Change 188: replace | old pp. [3] | new pp. [3]

Before (including deletions): geometry

After: geometry,

Change 189: replace | old pp. [3] | new pp. [3]

Before (including deletions): directional preference, and

After: we

Change 190: replace | old pp. [3] | new pp. [3]

Before (including deletions): a degree- budget completion:

After: degree-budget completion,

Change 191: replace | old pp. [3] | new pp. [3]

Before (including deletions): formulation [22], [23]

After: formula- tion [24], [25]

Change 192: replace | old pp. [3] | new pp. [3]

Before (including deletions): per-pin degree and thereby limits over-merge when endpoints are dropped or displaced. This differs from graph

After: the number of recovery connections per pin. Graph

Change 193: replace | old pp. [3] | new pp. [3]

Before (including deletions): engineering [24], which

After: engineer- ing [26]

Change 194: replace | old pp. [3] | new pp. [3]

Before (including deletions): gate-level

After: gate- level

Change 195: replace | old pp. [3] | new pp. [3]

Before (including deletions): a purely geometric model whose

After: geometric, and its

Change 196: replace | old pp. [3] | new pp. [3]

Before (including deletions): relationships

After: relations

Change 197: delete | old pp. [3] | new pp. []

Before (including deletions): wire

After: (none)

Change 198: delete | old pp. [3] | new pp. []

Before (including deletions): com- ponent

After: (none)

Change 199: delete | old pp. [3] | new pp. []

Before (including deletions): A complementary recent direction replaces rule- based connectivity with a learned model:

After: (none)

Change 200: replace | old pp. [3] | new pp. [3]

Before (including deletions): [25] pre- dicts

After: [27] predicts

Change 201: replace | old pp. [3] | new pp. [3]

Before (including deletions): schematics. Such

After: schematics; such learned

Change 202: replace | old pp. [3] | new pp. [3]

Before (including deletions): are powerful but require task-specific connec- tivity training data and, trained on printed schematics, do not transfer to hand-drawn images without retraining; our geometric model needs no

After: need

Change 203: replace | old pp. [3] | new pp. [3]

Before (including deletions): and is the focus here.

After: data, whereas our join needs none.

Change 204: replace | old pp. [3] | new pp. [3]

Before (including deletions): CircuitNet [26] likewise rely on

After: Cir- cuitNet [28] combine

Change 205: replace | old pp. [3] | new pp. [3]

Before (including deletions): plus

After: with

Change 206: delete | old pp. [3] | new pp. []

Before (including deletions): six-stage pipeline. The full pipeline proceeds through

After: (none)

Change 207: replace | old pp. [3] | new pp. [3]

Before (including deletions): detection via oriented bounding box

After: detection with an oriented-bounding-box

Change 208: replace | old pp. [3] | new pp. [3]

Before (including deletions): occlusion (filling detected regions)

After: occlusion, which fills each component region, enlarged by 15% of its box size (at least 5 px),

Change 209: replace | old pp. [3] | new pp. [3]

Before (including deletions): local

After: the

Change 210: replace | old pp. [3] | new pp. [3]

Before (including deletions): color);

After: colour of its surroundings;

Change 211: replace | old pp. [3] | new pp. [3]

Before (including deletions): adaptive binarization;

After: binarization ($k = 0.285$, window 67 px) inside the union of component boxes plus 10 px padding;

Change 212: replace | old pp. [3] | new pp. [3]

Before (including deletions): connected component

After: connected-component

Change 213: replace | old pp. [3] | new pp. [3]

Before (including deletions): with morphological post-processing;

After: after a 3×3 elliptical closing, discarding components under 28 px;

Change 214: replace | old pp. [3] | new pp. [3]

Before (including deletions): wire end- point extraction via

After: wire-segment and endpoint estimation by

Change 215: replace | old pp. [3] | new pp. [3]

Before (including deletions): on

After: of each connected component, followed by overlap deduplication (12■, 8 px) and an anchor filter with a 16 px

Change 216: replace | old pp. [3] | new pp. [3]

Before (including deletions): regions;

After: distance;

Change 217: delete | old pp. [3] | new pp. []

Before (including deletions): wire

After: (none)

Change 218: replace | old pp. [3] | new pp. [3]

Before (including deletions): via

After: with

Change 219: replace | old pp. [3] | new pp. [3]

Before (including deletions): endpoint-graph model with

After: endpoint graph and

Change 220: replace | old pp. [3] | new pp. [3]

Before (including deletions): is a netlist mapping

After: maps each

Change 221: replace | old pp. [3] | new pp. [3]

Before (including deletions): pins

After: pin

Change 222: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: an

Change 223: replace | old pp. [3] | new pp. [3]

Before (including deletions): nodes, suitable for SPICE netlist generation.

After: node.

Change 224: delete | old pp. [3] | new pp. []

Before (including deletions): representative

After: (none)

Change 225: delete | old pp. [3] | new pp. []

Before (including deletions): a set of

After: (none)

Change 226: replace | old pp. [3] | new pp. [3]

Before (including deletions): where

After: with

Change 227: replace | old pp. [3] | new pp. [3]

Before (including deletions): $\cup P$.

After: $\cup P$ and five edge types.

Change 228: replace | old pp. [3] | new pp. [3]

Before (including deletions): with weight 0. This ensures

After: so that

Change 229: replace | old pp. [3] | new pp. [3]

Before (including deletions): of a wire are always in

After: belong to

Change 230: delete | old pp. [3] | new pp. []

Before (including deletions): VOLUME 11, 2023 3

After: (none)

Change 231: replace | old pp. [4] | new pp. [3]

Before (including deletions): endpoint pairs (ei, ej)

After: endpoints ei, ej on different wires

Change 232: replace | old pp. [4] | new pp. [3]

Before (including deletions): edge with weight $|e_i - e_j|$.

After: an edge.

Change 233: replace | old pp. [4] | new pp. [3]

Before (including deletions): captures wire

After: rule connects

Change 234: replace | old pp. [4] | new pp. [3]

Before (including deletions): corners,

After: corners

Change 235: replace | old pp. [4] | new pp. [3]

Before (including deletions): For each

After: Each

Change 236: replace | old pp. [4] | new pp. [3]

Before (including deletions): e

After: connects to at most one pin. The component-first assignment described below chooses the pin. If no component is assigned, the endpoint takes the pin within $d \leq \tau_{\text{pin}}$ with the smallest score, where the score is $d[1 - \alpha \max(0, \cos \theta)]$ with $\alpha = 0.35$

Change 237: delete | old pp. [4] | new pp. []

Before (including deletions): pin p with $|e - p| \leq \tau_{\text{pin}}$, add edge with weight $|e - p| \cdot (1 - \alpha \cdot \cos \theta)$, where

After: (none)

Change 238: delete | old pp. [4] | new pp. []

Before (including deletions): is

After: (none)

Change 239: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: outward

Change 240: replace | old pp. [4] | new pp. [3]

Before (including deletions): vector, and α controls directional preference. This

After: vector (for $d \leq 10 - 6 \text{ px}$ the score

Change 241: replace | old pp. [4] | new pp. [3]

Before (including deletions): the critical

After: d). If neither step finds a pin, no

Change 242: replace | old pp. [4] | new pp. [3]

Before (including deletions): type: it connects wires to components.

After: is added.

Change 243: replace | old pp. [4] | new pp. [3]

Before (including deletions): each

After: an

Change 244: replace | old pp. [4] | new pp. [3]

Before (including deletions): wire

After: a different

Change 245: replace | old pp. [4] | new pp. [3]

Before (including deletions): $w_j = (c, d)$,

After: w_j , add an edge

Change 246: replace | old pp. [4] | new pp. [3]

Before (including deletions): $\text{dist}(e, w_j) \leq \tau$, add edge with weight $\text{dist}(e, w_j)$.

After: is at most τ and the normalized projection parameter lies in $[0.15, 0.85]$.

Change 247: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: rule

Change 248: replace | old pp. [4] | new pp. [3]

Before (including deletions): where a wire terminates on another wire's body.

After: away from segment ends.

Change 249: replace | old pp. [4] | new pp. [3]

Before (including deletions): each component

After: a

Change 250: replace | old pp. [4] | new pp. [3]

Before (including deletions): wire

After: a

Change 251: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: add an edge

Change 252: replace | old pp. [4] | new pp. [3, 4]

Before (including deletions): $\text{dist}(p, w_j) \leq \tau$

After: is at most VOLUME 11, 2023 3τ

Change 253: replace | old pp. [4] | new pp. [4]

Before (including deletions): lands mid-span (not at w_j 's endpoints, which are handled by edge type 3), add edge with weight $\text{dist}(p, w_j)$.

After: parameter lies in $[0.05, 0.95]$.

Change 254: replace | old pp. [4] | new pp. [4]

Before (including deletions): binds components that tap

After: rule captures pins tapped

Change 255: replace | old pp. [4] | new pp. [4]

Before (including deletions): passing rail along the wire's body rather than at its end. It recovers connectivity that endpoint-only binding misses, without the all-to-all over-merge of radius grabbing.

After: rail.

Change 256: replace | old pp. [4] | new pp. [4]

Before (including deletions): τ_{join} , τ_{pin} ,

After: are scale-relative with fixed pixel clamps: $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$

Change 257: delete | old pp. [4] | new pp. []

Before (including deletions): are scale-relative: τ

After: (none)

Change 258: replace | old pp. [4] | new pp. [4]

Before (including deletions): $k \cdot s$ where

After: $\min(20, \max(8, 0.20s))$, in pixels. Here

Change 259: replace | old pp. [4] | new pp. [4]

Before (including deletions): characteristic

After: median component-box diagonal of the image (the upper middle value for an even count), with a wire-length

Change 260: replace | old pp. [4] | new pp. [4]

Before (including deletions): (median

After: as fallback when

Change 261: replace | old pp. [4] | new pp. [4]

Before (including deletions): diagonal), so

After: geometry is unavailable. The multipliers and clamps are set by hand. Because s is one scalar per image, these three tolerances do not vary with component size inside

Change 262: replace | old pp. [4] | new pp. [4]

Before (including deletions): single parameter setting works across circuits of varying size.

After: drawing.

Change 263: delete | old pp. [4] | new pp. []

Before (including deletions): we use a two-step assignment. First, assign

After: (none)

Change 264: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: is first assigned

Change 265: replace | old pp. [4] | new pp. [4]

Before (including deletions): its

After: a component. Its distance to a component is 0 inside the component's OBB and otherwise the distance to the nearest OBB edge (axis-aligned box distance when OBB vertices are unavailable). The endpoint is assigned to the

Change 266: replace | old pp. [4] | new pp. [4]

Before (including deletions): by bounding-box distance (0 if inside the bbox, using a generous radius of

After: within

Change 267: replace | old pp. [4] | new pp. [4]

Before (including deletions): · diagonal)). Second, within

After: δc), where δc is that component's own box diagonal. Because this radius grows with each component's diagonal, endpoint-to- component binding follows the size of each component in addition to the image-level scale s . Within

Change 268: replace | old pp. [4] | new pp. [4]

Before (including deletions): component, select

After: component,

Change 269: delete | old pp. [4] | new pp. []

Before (including deletions): pin by geometric position: for horizontal components ($w > h$), left pin if

After: (none)

Change 270: replace | old pp. [4] | new pp. [4]

Before (including deletions): $x < \text{center } x$, else right pin; for vertical components ($h > w$), top pin if endpoint $y < \text{center } y$, else bottom pin. This handles

After: takes

Change 271: replace | old pp. [4] | new pp. [4]

Before (including deletions): common case where endpoints land inside a component's bounding box but far from its

After: nearest

Change 272: replace | old pp. [4] | new pp. [4]

Before (including deletions): our endpoint- graph model.

After: the endpoint graph.

Change 273: replace | old pp. [4] | new pp. [4]

Before (including deletions): computed as

After: the

Change 274: replace | old pp. [4] | new pp. [4]

Before (including deletions): projected

After: accumulated with union-find and projected

Change 275: replace | old pp. [4] | new pp. [4]

Before (including deletions): is a netlist mapping

After: maps

Change 276: replace | old pp. [4] | new pp. [4]

Before (including deletions): The endpoint-graph join may leave

After: After base joining,

Change 277: delete | old pp. [4] | new pp. []

Before (including deletions): component

After: (none)

Change 278: replace | old pp. [4] | new pp. [4]

Before (including deletions): “floating”: touching

After: lie in a net that contains

Change 279: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: pins of

Change 280: delete | old pp. [4] | new pp. []

Before (including deletions): component's net, with no wire connecting them to another

After: (none)

Change 281: replace | old pp. [4] | new pp. [4]

Before (including deletions): This indicates a dropped or over-displaced wire endpoint that the detector failed to place near the correct pin. Degree-budget completion identifies

After: Completion treats

Change 282: delete | old pp. [4] | new pp. []

Before (including deletions): and reconnects them via constrained optimization. A pin p is treated

After: (none)

Change 283: replace | old pp. [4] | new pp. [4]

Before (including deletions): floating if its net contains only pins from p's own component,

After: candidates for recovery,

Change 284: replace | old pp. [4] | new pp. [4]

Before (including deletions): such as gnd, vss,

After: (ground, supply markers,

Change 285: replace | old pp. [4] | new pp. [4]

Before (including deletions): antenna, junction,

After: antennas, probes, junctions

Change 286: replace | old pp. [4] | new pp. [4]

Before (including deletions): terminal. For

After: terminals). Completion keeps the base nets and proposes connections from

Change 287: replace | old pp. [4] | new pp. [4]

Before (including deletions): p, we generate candidate edges

After: p

Change 288: delete | old pp. [4] | new pp. []

Before (including deletions): all

After: (none)

Change 289: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: of other components

Change 290: replace | old pp. [4] | new pp. [4]

Before (including deletions): components within

After: nets. A candidate's distance must not exceed the

Change 291: replace | old pp. [4] | new pp. [4]

Before (including deletions): $p \cdot s$,

After: $p \tau c$,

Change 292: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: $\tau c = \min(60, \max(24, 0.62s)) p x$ and

Change 293: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: = 4. For each candidate we look for a wire witness: a detected segment whose endpoints lie near p and q in either orientation. The witness cost

Change 294: replace | old pp. [4] | new pp. [4]

Before (including deletions): reach factor; we use $p = 4$ for

After: smallest, over segments and orientations, of

Change 295: replace | old pp. [4] | new pp. [4]

Before (including deletions): deployed scale-relative join, and Section IV-D shows

After: larger endpoint-to-pin distance, bounded by r . Without

Change 296: replace | old pp. [4] | new pp. [4]

Before (including deletions): broad F1 plateau for $p \in [3, 5]$. Candidate edge weight

After: witness, the candidate cost

Change 297: replace | old pp. [4] | new pp. [4]

Before (including deletions): Euclidean distance $\|p - q\|$. We then solve

After: $\|p - q\|$. The candidates form

Change 298: replace | old pp. [4] | new pp. [4]

Before (including deletions): min-cost b-matching problem, $\min_{P(p,q) \in E'} w(p, q) \cdot x_{pq}$ subject to $P_{pq} x_{pq} \leq 1$ for each

After: rectangular assignment matrix with one row per

Change 299: replace | old pp. [4] | new pp. [4]

Before (including deletions): pin p , using the Hungarian algorithm [22] (`scipy.optimize.linear_sum_assignment`).

After: pin.

Change 300: replace | old pp. [4] | new pp. [4]

Before (including deletions): offers

After: supplies

Change 301: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: slot columns, limited by its candidate demand, and each row has a private dummy column of cost 1.5r for opting out. Unavailable assignments cost 106. The Hungarian algorithm [24] (`scipy.optimize.linear_sum_assignment`) assigns every row to one column, with at most one row per column; the target

Change 302: replace | old pp. [4] | new pp. [4]

Before (including deletions): so legitimate multi-terminal nets

After: implement bounded-capacity matching [25]. Matches

Change 303: replace | old pp. [4] | new pp. [4]

Before (including deletions): not starved,

After: applied in sequence. Dummy

Change 304: replace | old pp. [4] | new pp. [4]

Before (including deletions): a per-pin opt-out leaves a pin with

After: unavailable assignments add

Change 305: delete | old pp. [4] | new pp. []

Before (including deletions): good target unconnected rather than forcing a spurious

After: (none)

Change 306: replace | old pp. [4] | new pp. [4]

Before (including deletions): Finally, we reject any matched edge (p, q)

After: A merge is skipped

Change 307: replace | old pp. [4] | new pp. [4]

Before (including deletions): p and q belong to

After: both pins are already in

Change 308: replace | old pp. [4] | new pp. [4]

Before (including deletions): component

After: net

Change 309: replace | old pp. [4] | new pp. [4]

Before (including deletions): adding the edge would merge two

After: their

Change 310: delete | old pp. [4] | new pp. []

Before (including deletions): that

After: (none)

Change 311: replace | old pp. [4] | new pp. [4]

Before (including deletions): component, which structurally prevents short-circuiting two-terminal parts. Completion

After: component; this guard acts on completion merges only. Completion

Change 312: replace | old pp. [4] | new pp. [4]

Before (including deletions): are “wireless”: they represent inferred connections without

After: have no

Change 313: replace | old pp. [4] | new pp. [4]

Before (including deletions): segments, analogous to

After: segment, much like a

Change 314: replace | old pp. [4] | new pp. [4]

Before (including deletions): merges

After: merge

Change 315: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: a

Change 316: replace | old pp. [4] | new pp. [4]

Before (including deletions): editors.

After: editor.

Change 317: delete | old pp. [4] | new pp. []

Before (including deletions): completion

After: (none)

Change 318: replace | old pp. [4] | new pp. [4]

Before (including deletions): RESULTS

After: EXPERIMENTAL SETUP

Change 319: replace | old pp. [4] | new pp. [4]

Before (including deletions): SYNTHETIC EVALUATION FRAMEWORK

After: DATASETS AND IMAGE PROVENANCE All real images come from CGHD-1152 [1], a public corpus of hand-drawn circuit images organized by drafter.

Change 320: replace | old pp. [4] | new pp. [4]

Before (including deletions): author

After: use four evaluation sets. The wire benchmark has 134 images with human wire labels and annotated component boxes. The human-verified net benchmark has 31 images from 10 drafters. Thirty-four images were annotated and three were excluded before scoring (one unverified, two failing physical- sanity checks). Net proposals were bootstrapped by a conservative endpoint-graph join over the annotated wires and corrected by one annotator in a purpose-built verification tool. The 31 images contain 3–14 electrical components each (median 7);

Change 321: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: have at most five and 12 have at least ten. The independent held-out benchmark (164 images, 24 drafters) and the

Change 322: replace | old pp. [4] | new pp. [4]

Before (including deletions): circuits spanning 3–6

After: suite (15 authored circuits) are described in Sections V-D and V-A. The 134 wire-benchmark and 31 net-benchmark images are 704 × 704 copies from a Roboflow export of CGHD. The export resized each original without preserving its aspect ratio. Of the 31 net-benchmark images, 14 are also flipped or rotated relative to the CGHD file, and 16 (45 of the 134 wire-benchmark images) are CGHD’s binary stroke-segmentation renderings instead of photographs. Wire and net labels were drawn on these copies. The held-out benchmark uses the original CGHD photographs. B. CONNECTIVITY METRIC For each image we form the set of unordered pairs of electrical

Change 323: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: that share at least one net,

Change 324: replace | old pp. [4] | new pp. [4]

Before (including deletions): 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops,

After: compare predicted with reference pairs. The electrical subset contains resistors, capacitors, inductors, diodes, transistors, sources and integrated circuits; junctions, terminals, ground symbols and crossovers affect the score only through the connections they mediate. The metric ignores which terminal of a component carries a connection. Micro-F1 pools true positives (TP), false positives (FP) and false negatives (FN) over images, so a dense board contributes in proportion to its connections; it is our primary metric. Macro-F1, the mean of per-image F1, is reported alongside. C. STATISTICAL PROTOCOL All comparisons are paired over images. We report 95% percentile confidence intervals (CIs) from an image-level bootstrap with $B = 10,000$ resamples, and test each difference with three procedures: the bootstrap itself, a sign-flip permutation test that swaps the two methods' outputs within each image (exact when at most 22 images differ, otherwise 100,000 Monte Carlo draws),

Change 325: replace | old pp. [4] | new pp. [4]

Before (including deletions): 6- component ring (cyclic and bridge topologies being

After: Wilcoxon signed-rank test on per-image F1. The first two test

Change 326: replace | old pp. [4] | new pp. [4]

Before (including deletions): main stress cases for join robustness). Each circuit has a known ground-truth netlist and SPICE deck with a

After: micro-F1 difference; the Wilcoxon

Change 327: replace | old pp. [4] | new pp. [4]

Before (including deletions): voltage source. We inject five categories of error at four severity levels (L1–L4) on top of the clean (L0) baseline. These are Gaussian endpoint jitter with standard deviation σ (from 3 px at L1 to 14 px at L4), cut-short corruption in which

After: compares images. Holm's correction is applied within

Change 328: replace | old pp. [4] | new pp. [4]

Before (including deletions): wire is shortened inward by a fixed length to model endpoints that stop short of the pin (from 4 px at L1 to 22 px at L4), full wire drops with probability p_{drop} (up to 0.20 at L4), wrong- pin snaps with probability p_{wp} (from 0.02 at L2 to 0.10 at L4), and anchor displacement, where with probability p_{disp} one endpoint is moved by up to a_{max} pixels (from 30 px at L1 to 80 px at L4). We report join F1 as the component-connectivity F1 score, measuring which pairs of components share a net in the predicted versus ground-truth netlist. We report simulation accuracy as the percentage of random seeds for which every source current in the predicted SPICE netlist matches the clean oracle within 5%. B. WIRE DETECTION BENCHMARK We evaluate on 134 images from CGHD-1152 with ground- truth wire and component labels. Figure 5 and Table 1 summarize the top configurations across 36 tested variants. The pipeline achieves median F1 = 1.000

After: procedure

Change 329: replace | old pp. [4] | new pp. [4]

Before (including deletions): benchmark set, with 87%

After: comparisons

Change 330: replace | old pp. [4] | new pp. [4]

Before (including deletions): images scoring $F1 \geq 0.90$. Sauvola adaptive binarization dominates all alternatives: Otsu yields $F1 = 0.789$, adaptive Gaussian $F1 = 0.845$. Component occlusion matters: without it, component internals (text, symbols, internal wiring) generate hundreds of false wire detections.

After: a

Change 331: replace | old pp. [5] | new pp. [5]

Before (including deletions): approaches compared.

After: approaches.

Change 332: replace | old pp. [5] | new pp. [5]

Before (including deletions): grabs

After: takes

Change 333: replace | old pp. [5] | new pp. [5]

Before (including deletions): causing over-merging.

After: which over-merges.

Change 334: replace | old pp. [5] | new pp. [5]

Before (including deletions): Our endpoint-graph

After: Endpoint-graph

Change 335: replace | old pp. [5] | new pp. [5]

Before (including deletions): (wire body, endpoint–pin) with

After: and component-first pin assignment;

Change 336: replace | old pp. [5] | new pp. [5]

Before (including deletions): preference resolve connectivity without over-merging.

After: scoring applies only to the fallback pin search.

Change 337: replace | old pp. [5] | new pp. [5]

Before (including deletions): After

After: after

Change 338: replace | old pp. [5] | new pp. [5]

Before (including deletions): “floating” (touching

After: floating (connected

Change 339: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: to

Change 340: replace | old pp. [5] | new pp. [5]

Before (including deletions): Completion

After: completion

Change 341: replace | old pp. [5] | new pp. [5]

Before (including deletions): via

After: by

Change 342: replace | old pp. [5] | new pp. [5]

Before (including deletions): dashed), with self-loop

After: dashed); a

Change 343: replace | old pp. [5] | new pp. [5]

Before (including deletions): preventing shorts within two-terminal components. 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle Otsu adaptive Gaussian Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.928 0.949 0.950 0.959 0.973 0.976 FIGURE 5. Wire detection F1 across thresholding methods and pipeline variants (134 images). Our configuration (Sauvola binarization with

After: rejects merges between nets that already share

Change 344: replace | old pp. [5] | new pp. [5]

Before (including deletions): 16 px anchor filter) achieves F1 = 0.976. Otsu and triangle thresholding degrade significantly.

After: component.

Change 345: replace | old pp. [5] | new pp. [5]

Before (including deletions): Wire Detection Performance (134 images) Configuration

After: Synthetic join leaderboard: mean component-pair

Change 346: replace | old pp. [5] | new pp. [5]

Before (including deletions): Prec Rec Sauvola + anchor 16 px (ours) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola (w=61) + anchor 12 px 0.959 0.944 0.974 Sauvola (k=0.2875) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783 C. JOIN STRATEGY COMPARISON Table 2 and Figure 6 compare joining strategies on the syn- thetic evaluation framework (15

After: over 15

Change 347: replace | old pp. [5] | new pp. [5]

Before (including deletions): seeds, mean F1 over

After: seeds at each

Change 348: replace | old pp. [5] | new pp. [5]

Before (including deletions): levels L1–L4). The completion-based strategies are the most robust, main- taining F1 ≥ 0.94 at the highest error severity (L4), compared to F1 = 0.36 for the legacy radius union-find baseline, a 2.6 $\times$ TABLE 2. Join Strategy Leaderboard (Synthetic Eval)

After: level.

Change 349: replace | old pp. [5] | new pp. [5]

Before (including deletions): improvement. The radius join’s transitive grabbing causes catastrophic over-merging under noise:

After: table. We also count per-image wins, ties and losses (W/T/L). Unless stated otherwise (Section V-G), every join receives annotated component boxes and the pipeline’s own detected wires. V. RESULTS A. SYNTHETIC EVALUATION We authored 15 circuits with 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops and a 6-component ring. Each circuit has a known netlist, component values and a test voltage source. We inject five error types

Change 350: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: four severity levels (L1–L4) on top of the clean level L0: Gaussian endpoint jitter (σ from 3 px at L1 to 14 px at L4); cut-short wires, shortened inward by 4 px at L1 to 22 px at L4; wire drops with probability up to 0.20 at L4; wrong-pin snaps with probability from 0.02 at L2 to 0.10 at L4; and anchor displacement, which moves one endpoint by up to 30 px at L1 and 80 px at L4. Besides component-pair F1, we report simulation accuracy: the percentage of seeds for which every source current of the predicted netlist, simulated with the authored values, matches the clean reference within 5%. Table 1 and Figure 5 compare the join strategies. At

Change 351: delete | old pp. [5] | new pp. []

Before (including deletions): it shorts mul- tiple components onto shared nets. We emphasize, however, that

After: (none)

Change 352: replace | old pp. [5] | new pp. [5]

Before (including deletions): endpoint-graph base joins are themselves far more robust than

After: scale-relative graph with completion scores F1 0.95,

Change 353: replace | old pp. [5] | new pp. [5]

Before (including deletions): radius join, reaching F1 = 0.90

After: TABLE 2. Scale-relative completion join per synthetic circuit

Change 354: replace | old pp. [5] | new pp. [5]

Before (including deletions): (Table 2).

After: (16 seeds). Circuit Comps F1 Prec Sim. acc. Parallel R–R 3 1.00 1.00 25% R–R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode–R loop 3 0.99 1.00 69% Dual R loops 4 0.93 0.90 12% 4-component loop 4 0.94 0.91 56% Wheatstone bridge 6 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% rescue graph with completion 0.94, the rescue graph alone 0.90, the scale-relative graph alone 0.85 and radius union-find 0.36.

Change 355: replace | old pp. [5] | new pp. [5]

Before (including deletions): degree-budget completion adds

After: rescue graph differs from the scale-relative graph by

Change 356: replace | old pp. [5] | new pp. [5]

Before (including deletions): further +0.05 over

After: 12 px end-extension and a dead-end rescue step that gives dangling wire ends a longer directional reach. Table 2 breaks down our join by circuit at L4. Precision is at least 0.85 on every circuit. The Wheatstone bridge is

Change 357: replace | old pp. [5] | new pp. [5]

Before (including deletions): strongest

After: hardest case, with F1 0.82 and simulation accuracy 6%; the parallel R–R circuit keeps F1 1.00 at 25% simulation accuracy, so the two measures capture different properties. B. WIRE DETECTION Table 3 and Figure 6 summarize the best

Change 358: replace | old pp. [5] | new pp. [5]

Before (including deletions): these by recovering dropped connections

After: 36 extractor vari- ants on

Change 359: replace | old pp. [5] | new pp. [5]

Before (including deletions): base graph join leaves floating, while its per-pin edge budget keeps precision high. Pairing that completion

After: 134-image wire benchmark. Sauvola binarization with a 16 px anchor filter reaches F1 0.976 (median per-image F1 1.000; 87% of images at or above 0.90). Otsu thresholding reaches 0.789 and adaptive Gaussian thresholding 0.845. The configuration in the table deduplicates overlapping segments at 10■/18 px; the deployed 12■/8 px setting scores 0.973

Change 360: replace | old pp. [5] | new pp. [5]

Before (including deletions): high- precision scale-relative base (the scale-relative completion join used throughout) gives

After: same anchor filter. The score does not depend on image modality: F1 is 0.967 on

Change 361: replace | old pp. [5] | new pp. [5]

Before (including deletions): best synthetic robustness (0.95 at L4) and, more importantly,

After: 89 photographs and 0.986 on

Change 362: replace | old pp. [5] | new pp. [5]

Before (including deletions): best real-image connectivity (Section IV-D). The large gap is against the legacy radius join; against the strongest prior graph join the improvement is smaller but consistent across every severity level. For readability we omit two intermediate variants whose synthetic scores fall within rounding of rows shown (nearest-pin without anchoring, and the scale-relative base with end-extension but no dead-end rescue). D. REAL-IMAGE NET-LEVEL EVALUATION The synthetic results above measure robustness under a parametric error model. To validate on real detector output, we constructed, to our knowledge, the first human-verified net-

After: 45 stroke maps.

Change 363: replace | old pp. [6] | new pp. [6]

Before (including deletions): 6.

After: 5.

Change 364: replace | old pp. [6] | new pp. [6]

Before (including deletions): strategy robustness

After: strategies

Change 365: delete | old pp. [6] | new pp. []

Before (including deletions): bars give

After: (none)

Change 366: replace | old pp. [6] | new pp. [6]

Before (including deletions): connectivity

After: component-pair

Change 367: delete | old pp. [6] | new pp. []

Before (including deletions): The completion-based strategies (left) stay near 1.0, while the radius union-find and nearest-pin baselines (right) collapse at L3–L4.

After: (none)

Change 368: replace | old pp. [6] | new pp. [6]

Before (including deletions): connectivity

After: component-pair

Change 369: replace | old pp. [6] | new pp. [6]

Before (including deletions): Errors cause fragmentation rather than shorts (precision stays high;

After: TABLE 3. Wire detection on 134 images with annotated component boxes. Sauvola rows use 10■/18 px overlap deduplication. Configuration F1 Prec Rec Sauvola + anchor 16 px (ours) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola (w=61) + anchor 12 px 0.959 0.944 0.974 Sauvola (k=0.2875) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle (skeleton) Otsu adaptive Gaussian (skeleton) Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.845 0.949 0.950 0.959 0.973 0.976 FIGURE 6. Wire detection F1 across thresholding methods and pipeline variants (134 images, annotated component boxes). The adaptive Gaussian and Triangle bars use skeleton extraction (F1 = 0.845 and 0.758). C. HUMAN-VERIFIED BENCHMARK

Change 370: replace | old pp. [6] | new pp. [6]

Before (including deletions): 4);

After: 4 reports every join configuration and baseline on

Change 371: replace | old pp. [6] | new pp. [6]

Before (including deletions): balanced Wheatstone bridge is

After: 31-image human-verified benchmark, together with

Change 372: replace | old pp. [6] | new pp. [6]

Before (including deletions): hardest case. level ground truth for hand-drawn circuits: 31 images from CGHD-1152 in which

After: paired tests. Our join, degree-budget completion on the scale-relative base, reaches micro-F1 0.890 (P 0.919, R 0.864; macro-F1 0.901; TP/FP/FN 418/37/66; 95% CI [0.855, 0.924]). It beats

Change 373: replace | old pp. [6] | new pp. [6]

Before (including deletions): electrical net was traced and confirmed by a human annotator through a purpose-built verification tool. Because one of our comparison points is a vision-language model (Section IV-F), the human, not any model, is the verifier of record, so the answer key is independent of

After: deterministic comparator on

Change 374: replace | old pp. [6] | new pp. [6]

Before (including deletions): method under test. Net-level proposals were bootstrapped by a conservative endpoint-graph join

After: test after Holm correction. The margin

Change 375: delete | old pp. [6] | new pp. []

Before (including deletions): dataset's perfect wire labels and then corrected by hand; during verification the human consistently caught cases where the bootstrap join had wrongly split a single rail, and we additionally re-validate all conclusions on the independently authored synthetic ground truth to rule out residual bootstrap bias. Connectivity is scored as component-pair F1 restricted to SPICE-active components, identical to the synthetic metric. Table 3 reports every registered join strategy on the verified set, run on the pipeline's own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for reference. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 (P = 0.92, R = 0.86; 95% bootstrap CI [0.855, 0.924]), improving on the

After: (none)

Change 376: replace | old pp. [6] | new pp. [6]

Before (including deletions): default that

After: default, which

Change 377: replace | old pp. [6] | new pp. [6]

Before (including deletions): base (0.829)

After: base, is +0.061 [+0.025, +0.100] with 11 wins, 18 ties

Change 378: replace | old pp. [6] | new pp. [6]

Before (including deletions): the radius union-find join (0.667) (Figure 7). The gain over that rescue-based completion comes from the cleaner base: the rescue base's 12 px end-extension and dead-end rescue over-reach on noisy detected wires and cost precision, whereas the scale-relative base keeps precision high while the bounded completion still recovers recall. A reach sweep shows a broad F1 plateau across reach 3–5×s, so the setting is robust rather than tuned to a spike. Confidence intervals are wide at N = 31 and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 2). TABLE 3. Join Strategy and Baseline Comparison on Human-Verified Net-Level Ground Truth (31 real images, detected wires). Primary metric is micro-averaged (pair-pooled) component-pair F1; macro (mean per-image) F1 is shown for reference. Method F1 Prec Rec F1macro Deterministic joins (ours and ablations) Scale-rel. graph + completion (ours) 0.890 0.919 0.864 0.901 Rescue graph + completion (prior) 0.829 0.789 0.874 0.853 Scale-rel. graph (base) 0.816 0.929 0.727 0.838 Rescue graph (base) 0.787 0.811 0.764 0.813 Radius union-find (legacy) 0.667 0.698 0.638 0.674 Classical deterministic baselines (identical detected wires) Hough + proximity (best) 0.805 0.750 0.868 0.847 Connected-components (best) 0.624 0.965 0.461 0.611 Reference (non-deterministic / oracle wires) VLM (Claude Opus 4.8) 0.923 0.970 0.880 0.949 Scale-rel. + completion, perfect wires 0.890 0.913 0.868 0.916 This real-image ranking resolves an open question from prior work on this pipeline: against bootstrapped pilot labels the degree-budget completion had not consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787 for the rescue base alone), and the scale-relative variant is better still. The same ordering holds on the authored synthetic suite (Table 2), where the scale-relative completion join also ranks first, so the result is not an artifact of how the net-level labels were

bootstrapped. The dominant connectivity recipe in prior image-to-netlist systems erases component boxes and reads nets off the connected components, or contours, of the residual wire pixels. It is used by SINA [3], the concurrent contour-based 6 VOLUME 11, 2023

After: 2 losses.

Change 379: replace | old pp. [7] | new pp. [6]

Before (including deletions): connectivity

After: component-pair

Change 380: delete | old pp. [7] | new pp. []

Before (including deletions): net-level

After: (none)

Change 381: replace | old pp. [7] | new pp. [6]

Before (including deletions): (component-pair metric). Our scale-relative completion join leads all deterministic strategies

After: with annotated component boxes

Change 382: delete | old pp. [7] | new pp. []

Before (including deletions): the classical Hough and connected-component baselines (run on identical

After: (none)

Change 383: replace | old pp. [7] | new pp. [6]

Before (including deletions): wires). The dashed line marks a strong VLM (Claude Opus 4.8) evaluated on the same 31 images, from which our join is statistically indistinguishable (paired micro-F1 difference +0.033, 95% bootstrap CI [−0.009, +0.078]) while being deterministic, simulatable, and orders of magnitude cheaper.

After: wires.

Change 384: replace | old pp. [7] | new pp. [6]

Before (including deletions): Cls. node detector

After: Cls; the dashed line marks the VLM reference (Claude Opus 4.8). Completion lifts the scale-relative base from 0.816 to 0.890, mostly through recall (0.727 to 0.864). The scale-relative base is also more precise than the rescue base (0.929 versus 0.811), and completion keeps most

Change 385: replace | old pp. [7] | new pp. [6]

Before (including deletions): Peker et al. [5], AMSnet [13],

After: that precision (0.919, against 0.789 for completion on the rescue base). Leaving out any one image moves our micro-F1 within 0.885–0.900,

Change 386: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: no deletion changes

Change 387: replace | old pp. [7] | new pp. [6]

Before (including deletions): Bayer instance-segmentation pipeline [6],

After: sign of any comparison. A sweep of the completion reach over $p \in [3, 5]$ gives macro-F1 0.895–0.903. Several image-to-netlist systems erase component regions

Change 388: replace | old pp. [7] | new pp. [6]

Before (including deletions): was designed for clean printed schematics.

After: read nets from the remaining wire pixels with connected components or contours [3], [5].

Change 389: replace | old pp. [7] | new pp. [6]

Before (including deletions): hand-drawn photographs it has no precise operating point: pencil-stroke fragmentation splits each net into many components ($F1 < 0.25$ at sensible morphological gap-bridging), and only pathological closing that effectively merges by proximity lifts recall, at a steep precision cost. For a controlled comparison that isolates

After: our detected wires,

Change 390: delete | old pp. [7] | new pp. []

Before (including deletions): join algorithm from binarization quality, we run connected- component labeling on our own detected wire segments, identical input to our join, and its

After: (none)

Change 391: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: connected-component

Change 392: delete | old pp. [7] | new pp. []

Before (including deletions): only

After: (none)

Change 393: delete | old pp. [7] | new pp. []

Before (including deletions): =

After: (none)

Change 394: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: at precision 0.965: it rarely merges wrongly, but every stroke gap splits a net

Change 395: replace | old pp. [7] | new pp. [6]

Before (including deletions): collapses to 0.46: the blobs short or drop most nets), versus 0.890 for the endpoint-graph join (Table 3). A stronger classical alternative, Hough-transform line detection [18] followed by proximity linking, fares better, since

After: 0.461). Hough

Change 396: replace | old pp. [7] | new pp. [6]

Before (including deletions): denoises the fragmented strokes, and

After: [20] with proximity linking

Change 397: delete | old pp. [7] | new pp. []

Before (including deletions): micro-F1 =

After: (none)

Change 398: replace | old pp. [7] | new pp. [6]

Before (including deletions): best-tuned

After: best

Change 399: replace | old pp. [7] | new pp. [6]

Before (including deletions): tolerance, but still trails our join by 0.085 at markedly lower precision. Each baseline is reported at its

After: tolerance. Both baselines use the

Change 400: replace | old pp. [7] | new pp. [6]

Before (including deletions): over a tolerance sweep, so

After: of their sweep on

Change 401: replace | old pp. [7] | new pp. [6, 7]

Before (including deletions): are upper bounds

After: 31 images, which favours them. 6 VOLUME 11, 2023 TABLE 4. Component-pair connectivity

Change 402: replace | old pp. [7] | new pp. [7]

Before (including deletions): classical recipes. Explicit typed-endpoint reasoning

After: 31-image human-verified benchmark (annotated component boxes; detected wires unless noted). F1, P and R are micro-averaged. ■ is the micro-F1 of ours minus the comparator

Change 403: replace | old pp. [7] | new pp. [7]

Before (including deletions): bounded

After: 95% paired bootstrap CI; Holm-adjusted p values are given for the bootstrap, the permutation test on micro-F1 and the Wilcoxon test on per-image F1; W/T/L counts per-image wins, ties and losses from our side. Method F1 P R F1macro Δ[95% CI] Holm p (boot / perm / Wilcoxon) W/T/L Endpoint-graph joins Scale-rel. graph +

Change 404: replace | old pp. [7] | new pp. [7]

Before (including deletions): thus outperforms both blob/contour connectivity and line- primitive

After: (ours) 0.890 0.919 0.864 0.901 – – – Rescue graph + completion 0.829 0.789 0.874 0.853 +0.061 [+0.025, +0.100] <0.001 / 0.001 / 0.009 11/18/2 Scale-rel. graph (base) 0.816 0.929 0.727 0.838 +0.075 [+0.038, +0.113] <0.001 / 0.002 / 0.005 14/16/1 Rescue graph (base) 0.787 0.811 0.764 0.813 +0.103 [+0.065, +0.142] <0.001 / <0.001 / 0.002 17/14/0 Baselines on the same detected wires Radius union-find (legacy) 0.667 0.698 0.638 0.674 +0.224 [+0.166, +0.281] <0.001 / <0.001 / <0.001 25/6/0 Hough +

Change 405: replace | old pp. [7] | new pp. [7]

Before (including deletions): tracing.

After: (best) 0.805 0.750 0.868 0.847 +0.086 [+0.045, +0.127] <0.001 / 0.002 / 0.030 19/8/4 Connected components (best) 0.624 0.965 0.461 0.611 +0.267 [+0.136, +0.409] <0.001 / <0.001 / <0.001 21/8/2 References Ours with annotated wires 0.890 0.913 0.868 0.916 – – – VLM (Claude Opus 4.8) 0.923 0.970 0.880 0.949 –0.033 [–0.078, +0.008] 0.12 / 0.15 / 0.030 7/11/13

Change 406: replace | old pp. [7] | new pp. [7]

Before (including deletions): additionally attempted

After: also tried

Change 407: replace | old pp. [7] | new pp. [7]

Before (including deletions): image-to- netlist

After: image-to-netlist

Change 408: delete | old pp. [7] | new pp. []

Before (including deletions): end-to-end

After: (none)

Change 409: replace | old pp. [7] | new pp. [7]

Before (including deletions): our hand-drawn benchmark, and none could be fairly applied.

After: the benchmark.

Change 410: replace | old pp. [7] | new pp. [7]

Before (including deletions): classical printed- schematic

After: printed-schematic

Change 411: delete | old pp. [7] | new pp. []

Before (including deletions): installs and

After: (none)

Change 412: replace | old pp. [7] | new pp. [7]

Before (including deletions): connectivity, a domain- transfer failure.

After: con- nectivity.

Change 413: replace | old pp. [7] | new pp. [7]

Before (including deletions): [25]

After: [27]

Change 414: delete | old pp. [7] | new pp. []

Before (including deletions): which we ran end-to-end with the released model,

After: (none)

Change 415: replace | old pp. [7] | new pp. [7]

Before (including deletions): active

After: released

Change 416: replace | old pp. [7] | new pp. [7]

Before (including deletions): inductor,

After: inductor

Change 417: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: in

Change 418: replace | old pp. [7] | new pp. [7]

Before (including deletions): circuits contain

After: circuits,

Change 419: replace | old pp. [7] | new pp. [7]

Before (including deletions): appearance, so it cannot repre- sent hand-drawn discrete circuits.

After: drawings.

Change 420: replace | old pp. [7] | new pp. [7]

Before (including deletions): equiv- alent in spirit to

After: which

Change 421: replace | old pp. [7] | new pp. [7]

Before (including deletions): experiment in Section IV-F, rather than a deterministic algorithm. SINA's [3] anonymized

After: reference (Section V-I) covers. The

Change 422: replace | old pp. [7] | new pp. [7]

Before (including deletions): was unextractable

After: of SINA [3] could not be extracted,

Change 423: replace | old pp. [7] | new pp. [7]

Before (including deletions): [26]

After: [28]

Change 424: delete | old pp. [7] | new pp. []

Before (including deletions): only Co- lab

After: (none)

Change 425: replace | old pp. [7] | new pp. [7]

Before (including deletions): released

After: trained

Change 426: replace | old pp. [7] | new pp. [7]

Before (including deletions): This survey is itself informative: no prior system provides a deterministic, hand- drawn-ready connectivity module, so the controlled

After: The

Change 427: replace | old pp. [7] | new pp. [7]

Before (including deletions): above, both run

After: therefore stand in for these approaches

Change 428: delete | old pp. [7] | new pp. []

Before (including deletions): our

After: (none)

Change 429: replace | old pp. [7] | new pp. [7]

Before (including deletions): detected wires, are

After: inputs. Replacing

Change 430: delete | old pp. [7] | new pp. []

Before (including deletions): fair points of comparison, and our join exceeds both. Running the best join on the dataset's perfect wire labels rather than

After: (none)

Change 431: replace | old pp. [7] | new pp. [7]

Before (including deletions): leaves

After: with the annotated wires, with boxes held fixed, changes

Change 432: replace | old pp. [7] | new pp. [7]

Before (including deletions): unchanged at 0.890, while raising

After: from 0.8903 to 0.8898 and

Change 433: delete | old pp. [7] | new pp. []

Before (including deletions): only

After: (none)

Change 434: replace | old pp. [7] | new pp. [7]

Before (including deletions): 0.916 (Table 3), so

After: 0.916. On this benchmark,

Change 435: replace | old pp. [7] | new pp. [7]

Before (including deletions): wire detector accounts

After: extracted wires are as good as the annotated ones

Change 436: replace | old pp. [7] | new pp. [7]

Before (including deletions): essentially zero pair-weighted F1.

After: joining.

Change 437: replace | old pp. [7] | new pp. [7]

Before (including deletions): residual error

After: join also scores alike on both image modalities: micro-F1

Change 438: replace | old pp. [7] | new pp. [7]

Before (including deletions): intrinsic

After: 0.883 on the 15 photographs and 0.895 on the 16 stroke maps. D. INDEPENDENT HELD-OUT BENCHMARK The 31-image benchmark has three weaknesses: it is small, its labels were bootstrapped from a graph

Change 439: replace | old pp. [7] | new pp. [7]

Before (including deletions): ambiguity

After: similar to ours, and our configuration was chosen

Change 440: replace | old pp. [7] | new pp. [7]

Before (including deletions): hand-drawn geometry,

After: it. We therefore built a second benchmark whose labels come from the CGHD authors' own annotations and whose images were

Change 441: replace | old pp. [7] | new pp. [7]

Before (including deletions): wire detection, consistent

After: used to develop or tune the join. Reference construction. CGHD provides binary stroke-segmentation maps and symbol instance polygons for a subset of its images. Extending the dataset's own wire-generation tooling, we subtract the dilated symbol polygons from the stroke map and take the remaining connected components as conductors. A conductor connects every polygon it touches, except text; junction polygons merge the conductors they touch. At a crossover polygon, the arms are paired straight through by their local direction ($180^\circ \pm 45^\circ$), and a crossover that cannot be paired is flagged as ambiguous. Composite out- TABLE 5. Independent held-out benchmark (164 CGHD photographs, 24 drafters, 3874 reference pairs; annotated component boxes). ■ is ours minus the comparator in micro-F1, with 95% paired bootstrap CI. Every difference has Holm-adjusted $p = 0.0006$, the resolution floor of the tests. Method F1 P R Δ [95% CI] W/T/L Ours 0.711 0.766 0.664 – – Rescue + compl. 0.659 0.622 0.699 +0.053 [0.033, 0.074] 85/44/35 Scale-rel. base 0.595 0.785 0.479 +0.117 [0.097, 0.138] 112/46/6 Rescue base 0.592 0.631 0.558 +0.119 [0.097, 0.142] 113/31/20 Radius u-f 0.518 0.478 0.565 +0.194 [0.159, 0.229] 136/13/15 Hough 0.480 0.328 0.895 +0.231 [0.168, 0.282] 90/22/52 Conn. comp. 0.594 0.798 0.473 +0.117 [0.083, 0.156] 117/12/35 lines (transformers, optocouplers and relays that contain sub-parts) group their parts but take no contacts. Of 257 images, 179 pass all quality checks; the others have an ambiguous crossover (29), an isolated part (17) or an over-connected part (28). Validation. On the 17 images shared

Change 442: replace | old pp. [7] | new pp. [7]

Before (including deletions): 0.976 wire-detection F1 reported above (Table 1). The join degrades only gently

After: human-verified benchmark, the reference agrees

Change 443: replace | old pp. [7] | new pp. [7]

Before (including deletions): circuit size, with per-image F1 falling from 0.94

After: the human nets at micro-F1 0.939 (P 0.993, R 0.891; exact

Change 444: replace | old pp. [7] | new pp. [7]

Before (including deletions): circuits with ≤ 6 components to 0.86 above that and only weak correlation $r = -0.19$, and remains high-precision throughout, in contrast to the VLM, whose precision collapses

After: 11 of 17; 0.964

Change 445: replace | old pp. [7] | new pp. [7]

Before (including deletions): densest circuits (Section IV-F). E. PER-CIRCUIT ANALYSIS Table 4 shows

After: 15 clean images). We inspected every dis- agreement. Most are pairs

Change 446: replace | old pp. [7] | new pp. [7]

Before (including deletions): scale-relative completion join's perfor- mance across eight representative circuits

After: reference misses because of stroke gaps or ambiguous crossings, and seven follow

Change 447: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: a convention:

Change 448: replace | old pp. [7] | new pp. [7]

Before (including deletions): 15-circuit synthetic corpus at L4 (extreme error injection, 16 seeds). TABLE 4. Scale-Relative Completion Join Performance per Circuit at L4 (16 seeds) Circuit Comps F1 Prec Sim. acc. Parallel R-R 3 1.00 1.00 25% R-R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode-R loop 3 0.99 1.00 69% Dual R loops 4 0.93 0.90 12% 4-component loop 4 0.94 0.91 56% Wheatstone

After: human nets

Change 449: replace | old pp. [7] | new pp. [7]

Before (including deletions): 6 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% Precision remains high (≥ 0.85) across

After: switches. In five pairs, the reference was right and the human labels wrong. The ref- erence mainly under-connects, so it understates the precision of every method. We fixed the quality filter and a junction- handling rule after inspecting these 17 overlaps. Protocol. The held-out set contains the 164 clean images outside the 31, with 3874 reference pairs from

Change 450: replace | old pp. [7] | new pp. [7]

Before (including deletions): circuits even at L4: errors cause fragmentation (dropped connections) rather than shorts (false merges).

After: 24 drafters.

Change 451: replace | old pp. [7] | new pp. [7]

Before (including deletions): Wheatstone bridge

After: input

Change 452: replace | old pp. [7] | new pp. [7]

Before (including deletions): hard- est topology (F1 = 0.82):

After: CGHD photograph with

Change 453: replace | old pp. [7] | new pp. [7]

Before (including deletions): balanced arms make

After: aspect ratio preserved and its long side resized to 1024 px. We chose this resolution on wire-level labels of

Change 454: replace | old pp. [7] | new pp. [7]

Before (including deletions): inferred node structure sensitive to a single mis-placed connection,

After: 52 wire-benchmark overlaps, before computing any join score. Results. Table 5 gives the results. Our join reaches micro-F1 0.711 [0.671, 0.748] (P 0.766, R 0.664; macro-F1 0.724)

Change 455: replace | old pp. [7] | new pp. [7]

Before (including deletions): simulation accuracy falls to 6%. F. VLM CONNECTIVITY EXPERIMENT To quantify the separation argued in Section II, that connectivity is better solved by an explicit geometric model than left to a VLM, we ran a first-party experiment with a strong general VLM (Claude Opus 4.8). To make the comparison fair, the VLM is given exactly what the join receives: the scanned image

After: beats every comparator,

Change 456: replace | old pp. [7] | new pp. [7]

Before (including deletions): ground-truth component boxes

After: ordering of the endpoint-graph variants as on the human-verified set. Hough

Change 457: replace | old pp. [8] | new pp. [8]

Before (including deletions): labelled

After: linking keeps recall high (0.895) but merges many nets (precision 0.328); the best Hough configuration tuned on this set itself reaches 0.509. The result holds under three changes to the evaluation set: excluding every other photograph of the 31 benchmark drawings (155 images) gives 0.707, scoring all 257 images including the unclear ones gives 0.626, and a reference variant that treats switches as closed gives 0.698. Across the 24 drafters, our micro-F1 ranges from 0.465 to 0.917 (median 0.709, interquartile range 0.658–0.770), and our join is the best of the evaluated methods for 15 drafters. A human audit of 40 randomly sampled held-out images [RESULT PENDING]. Gap to the human-verified benchmark. The drop from 0.890 to 0.711 comes mainly from the input images. The held-out photographs belong to the CGHD subset for which the dataset provides stroke maps, and for this subset the 704×704 benchmark copies are mostly those stroke maps (16 of the 17 overlaps). On the photographs, wire F1 falls to about 0.80–0.82, against 0.98 on the benchmark copies. On the 17 overlapping images, our join scores 0.896 on the benchmark copies and 0.728 on the photographs at long side 1024. Re-weighting the held-out images to the circuit-size distribution of the 31 raises our score

Change 458: replace | old pp. [8] | new pp. [8]

Before (including deletions): index,

After: about 0.01,

Change 459: insert | old pp. [] | new pp. [8]

Before (including deletions): (none)

After: scoring against the human nets instead of the reference changes it by about 0.01. The remaining gap of about 0.16

Change 460: replace | old pp. [8] | new pp. [8]

Before (including deletions): asked

After: due

Change 461: replace | old pp. [8] | new pp. [8, 9, 10]

Before (including deletions): return,

After: input quality: on these photographs, wire extraction is the dominant source of error, while the ranking of the joins carries over. E. ROBUSTNESS 1) Circuit complexity Table 6 splits both benchmarks by the number of electrical components. On the 31 images, F1 falls with complexity for every method, including the VLM: across images, our F1 correlates with component count at Spearman $\rho = -0.53$ ($p = 0.002$). Our recall drop from the ≤ 5 to the ≥ 10 stratum, -0.068 [-0.18 , $+0.05$] (permutation $p = 0.32$), is not significant. On images with at least ten components, our join beats the rescue-graph completion, the scale-relative base and Hough by $+0.078$ to $+0.106$, with every CI excluding zero, and trails the VLM by -0.026 [-0.087 , $+0.030$]. On the held-out benchmark, micro-F1 falls from 0.845 for at most five components to 0.693 for 20 or more, while

Hough falls to 0.347. 2) Component-size variation within an image A single scale s per image could fail when component sizes vary within a drawing. On the 31 images, the overall dispersion of component sizes is unrelated to F1 ($\rho = +0.18$, $p = 0.33$). The ratio of the largest to the smallest electrical-component diagonal correlates with F1 at $\rho = -0.60$, but it also grows with component count ($\rho = 0.62$); controlling for count, the partial correlation is -0.41 ($p = 0.024$) for our join and -0.34 for the VLM. The size of the largest component relative to s has partial correlation near zero with F1. The three images that combine an IC with small discrete parts score 0.854 against 0.898 on the other 28. TABLE 6. Micro-F1 by number of electrical components per image. The VLM was run on the human-verified benchmark only. Components Images Ours Rescue + compl. Hough VLM Human-verified benchmark (31 images) ≤ 5 15 0.940 0.933 0.892 0.982 6–9 4 0.846 0.817 0.843 0.902 ≥ 10 12 0.886 0.808 0.779 0.912 Held-out benchmark (164 images) ≤ 5 41 0.845 0.856 0.835 – 6–9 41 0.779 0.756 0.701 – 10–19 60 0.694 0.656 0.593 – ≥ 20 22 0.693 0.608 0.347 – 3) Controlled rescaling We resampled all 31 images by factors from 0.35 to 3 and re-projected the annotated boxes and wires (Fig. 8). With annotated wires, our join stays within ± 0.005 of its native score from $0.5\times$ to $3\times$, with every CI including zero. It drops only at $0.35\times$ (-0.035 [-0.061 , -0.005]), where the lower pixel clamps bind for every image. With annotated wires, radius union-find with its fixed 30 px radius falls from 0.690 at native scale to 0.429 at $2\times$ and 0.153 at $3\times$. With wires re-extracted from each rescaled image, the full pipeline scores 0.580, 0.690, 0.811, 0.890, 0.877, 0.866 and 0.866 at 0.35, 0.5, 0.75, 1, 1.5, 2 and $3\times$. Upscaling costs at most 0.024 (CIs include zero). The loss under downscaling comes from wire extraction, whose Sauvola window, minimum component area and occlusion margins are set in pixels: at $0.35\times$ the extractor finds 241 of 675 annotated segments. Completion recovers much of that loss; the scale-relative base alone scores 0.21 at $0.35\times$. To separate the contribution of the scale-relative tolerances, we ran the same algorithm with fixed pixel tolerances (τ_{pin} , τ_{join} , τ_{t} of 30, 14 and 10 px, completion reach 120 px). It stays within 0.016 of our join between $0.5\times$ and $2\times$. The scale-relative version wins only where the fixed values are most mismatched: by $+0.015$ with annotated wires at $0.5\times$ and by $+0.033$ in the full pipeline at $3\times$, both with CIs excluding zero. Most of the scale robustness therefore comes from component-relative endpoint assignment and completion; the scalar tolerances $\tau = k \cdot s$ contribute a smaller share. The clamps are tuned to native resolution: $k \cdot s$ falls outside a clamp for 39% of images at native scale, and removing the clamps costs 0.023 in the full pipeline at native scale. 4) Synthetic mixed-size circuits We redrew one component per synthetic circuit at 0.5, 2 or 3 times its size, keeping only geometrically clean variants, and applied the L1–L4 error model. Enlarging one component costs our join about 0.005 in mean F1, and it stays about 0.03 above the fixed-pixel variant. Shrinking one component costs 0.012. The failure case is a Wheatstone bridge with one 40×15 px arm: in all five variants, the scale-relative methods and radius union-find add one spurious pair even on clean wires, while the fixed-pixel graphs stay exact. A component 8 VOLUME 11, 2023 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (a) Join only: annotated wires 0.35 0.5 0.75 1 1.5 2 3 Image scale factor f (log scale; native = 1) 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (b) Full: wires re-extracted 95% bootstrap CI Scale-rel. graph + completion Fixed-px graph + completion Scale-rel. graph (base) Radius union-find (30 px) FIGURE 8. Component-pair micro-F1 on the 31 images resampled by factor f (native $f = 1$). (a) Join only, with annotated wires rescaled with the image. (b) Full pipeline, with wires re-extracted from each rescaled image. Components are annotated boxes in both panels; shaded bands are 95% bootstrap CIs for our join. TABLE 7. Edge-rule ablation on the 31-image benchmark, for the base graph and the full pipeline with completion. Base graph Full pipeline Configuration F1 TP/FP/FN F1 TP/FP/FN All edge types (ours) 0.816 352/27/132 0.890 418/37/66 No T-junctions 0.816 352/27/132 0.890 418/37/66 No rail taps 0.816 352/27/132 0.890 418/37/66 No T-junctions, no rail taps 0.816 352/27/132 0.890 418/37/66 No directional preference 0.816 352/27/132 0.890 418/37/66 Fixed-pixel base tolerances 0.820 356/28/128 0.890 418/37/66 much smaller than the median is thus the weak point of a median-derived tolerance. F. ABLATIONS Table 7 removes individual edge rules from the base graph, with and without completion. Disabling T-junctions, rail taps or directional preference leaves the aggregate base-graph counts unchanged, so these rules give no measurable gain on this benchmark. Fixed-pixel base tolerances raise base micro-F1 from 0.816 to 0.820. With completion, every row ties at TP/FP/FN 418/37/66. The rescaling and mixed-size experiments (Section V-E) test the tolerance design under conditions that this benchmark does not cover. Table 8 tests the remaining mechanisms. Removing only the occlusion step, with cropping and all extractor settings TABLE 8. Mechanism ablations on the 31-image benchmark. “Changed” counts images whose predicted pairs differ from the reference arm. Arm TP/FP/FN F1 F1macro Changed Occlusion on (ours) 418/37/66 0.890 0.901 – Occlusion off 103/26/381 0.336 0.294 31/31 Guard on (ours) 418/37/66 0.890 0.901 – Guard off 418/37/66 0.890 0.901 0/31 Witness required 413/37/71 0.884 0.907 6/31 TABLE 9. End-to-end joining with detected components on the 31-image benchmark. The lower block replaces annotated boxes with detector output one factor at a time (confidence 0.5); ■ is the change from the previous row. Components F1 Prec Rec TP/FP/FN Annotated boxes 0.890 0.919 0.864 418/37/66 Detector, conf 0.25 0.613 0.575 0.655 317/234/167 Detector, conf 0.35 0.616 0.582 0.655 317/228/167 Detector, conf 0.50 0.627 0.603 0.653 316/208/168 Sequential step F1 Δ TP/FP/FN Drop missed components 0.759 -0.131 329/54/155 Add spurious detections 0.687 -0.072 322/131/162 Use detected geometry 0.644 -0.043 329/208/155 Use detected classes 0.627 -0.017 316/208/168 held fixed, exposes component pixels to the extractor, which then returns 466 wire segments instead of 666; micro-F1 falls from 0.890 to 0.336 ($P = 0.798$, $R = 0.213$), and predictions change on all 31 images. The shared-component guard fires on 18 of 31 images and rejects 38 candidate merges. These merges would join two pins of the same component, which the component-pair metric cannot see, so the scores are unchanged. Requiring a wire witness for every completion edge changes six images and lowers micro-F1 to 0.884 while raising macro-F1 to 0.907. The rescaling experiment adds two further arms, fixed-pixel tolerances and unclamped tolerances (Section V-E), and Section V-G decomposes the loss from detected components. G. END-TO-END WITH DETECTED COMPONENTS We replaced the annotated boxes with the outputs of our trained component detector, a YOLO26m-OBb model with 16 merged classes that reaches mAP@0.5 of 89.0% on its validation split, with its lowest class recall (70.7%) on crossovers. The detector runs on the EXIF-corrected CGHD original at its training size (long side 1024 px), and its boxes are mapped into the

benchmark frame. We use the deployment confidence threshold of 0.5 without tuning. On the electrical components, detection precision, recall and F1 are 0.886, 0.858 and 0.872, and the same when the class must also match. As a check of the scoring path, feeding the annotated boxes through the same mapping and matching code reproduces 418/37/66. Table 9 gives the results. End-to-end micro-F1 is 0.627 [0.519, 0.731] (macro-F1 0.656), a change of -0.263 [-0.366 , -0.170] from annotated boxes; the lower threshold VOLUME 11, 2023 9 olds give 0.613 and 0.616. Replacing the annotated boxes one factor at a time attributes -0.131 to missed electrical components, -0.072 to spurious detections, -0.043 to box localization and -0.017 to classification; with a sequential decomposition, the shares depend on the order of the steps. Missed components cost the most because every pair involving a missed component becomes a false negative. The detector is sensitive to orientation (its training used rotation augmentation of at most $\pm 10^\circ$). On the 14 images whose benchmark copy is re-oriented relative to the CGHD file, end-to-end micro-F1 is 0.491, against 0.739 on the other 17. H. CROSSOVERS Edge types 2, 4 and 5 do not consult crossover labels, so a missed crossover could merge two independent wires. The 31 images contain 13 annotated crossovers in eight images. With annotated boxes, recall on these eight images is 0.834, against 0.884 on the 23 without a crossover ($p = 0.51$). In C242_D1_P1, suppressing the two endpoint–endpoint edges adjacent to crossings changes TP/FP/FN from 27/4/0 to 19/0/8: the edges that create the false pairs also carry true connections. Deleting the annotated crossover boxes before joining changes predictions on 6 of 31 images, leaves the net partitions identical on 23, and changes micro-F1 by $+0.010$. In the end-to-end run, the detector finds all 13 crossovers and labels each

Change 462: replace | old pp. [8] | new pp. [10]

Before (including deletions): JSON,

After: a crossover at every tested threshold, so none of

Change 463: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: false pairs in Table 9 comes from a missed or misclassified crossover. Removing the correctly detected crossovers lowers end-to-end micro-F1 by 0.019. Thirteen instances are too few to test the 70.7% validation recall. I. VLM REFERENCE We ran Claude Opus 4.8 as a reference system. It receives the same annotated component boxes as our join, drawn on the image and numbered, and neither method receives annotated wires. The model returns JSON

Change 464: replace | old pp. [8] | new pp. [10]

Before (including deletions): we

After: each image is a separate call with no shared context. We

Change 465: replace | old pp. [8] | new pp. [10]

Before (including deletions): the resulting component-pairs

After: its output

Change 466: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: component-pair

Change 467: replace | old pp. [8] | new pp. [10]

Before (including deletions): Each

After: The VLM reaches micro-F1 0.923 (P 0.970, R 0.880; macro-F1 0.949) and is exact on 21 of 31 images, against 11 for our join. On pooled pairs, our join trails by -0.033 [-0.078 , $+0.008$], which is not significant (Holm-adjusted bootstrap $p = 0.12$, permutation $p = 0.15$; Table 4). Per image, the VLM is better: it wins 13 images, ties 11 and loses 7 (Wilcoxon $p = 0.030$). Its errors follow the same complexity trend as ours (Table 6). The two systems differ in what they require: the VLM needs a call to a proprietary API for every

Change 468: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: and returns no geometric trace of its decisions, whereas our join runs locally and deterministically without connectivity training data, and every recovered connection maps to a wire segment or a completion edge. J. PER-DRAFTER RESULTS Table 10 groups the 31-image results by CGHD drafter, using the drafter paths of the published CGHD-1152 file index, TABLE 10. Micro-F1 of our join by CGHD drafter on the 31-image benchmark (drafters with at least four images). Drafter Images F1

F1macro TP/FP/FN drafter_1 7 0.933 0.907 56/6/2 drafter_2 5 0.854 0.874 85/14/15 drafter_10 5 0.987 0.995 37/1/0 drafter_12 4 0.913 0.938 21/0/4 All 31 0.890 0.901 418/37/66 which place every benchmark filename under exactly one drafter. The four drafters with at least four images range from micro-F1 0.854 to 0.987. The held-out benchmark gives the spread over all 24 drafters (Section V-D). VI. DISCUSSION The endpoint-graph join with completion

Change 469: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: the strongest deterministic method on both real benchmarks, and its margins are significant on both, although the benchmarks differ in labels, images and drafters. The held-out benchmark addresses the three main weaknesses of the human-verified set: its labels come from the dataset authors' annotations and not from a join, it covers every drafter, and none of its images was used to develop the method. Completion accounts for most of the gain over the base graphs, and on the high-precision scale- relative base it adds recall while keeping precision above 0.9. The experiments also locate the remaining errors. With annotated components on clean copies, extracted wires are as good as annotated ones. On difficult photographs, wire extraction limits accuracy; with detected components, missed detections do. Both suggest concrete next steps: an extractor whose pixel parameters follow the component scale, resampling inputs to a nominal component size, and a detector trained with full rotation augmentation. The VLM reference shows that a general model given component boxes can exceed a geometric join per image on this task. The geometric join remains useful where a local, deterministic and inspectable result is required. A. SCOPE AND LIMITATIONS Task and metric. The pipeline recovers structural connectivity. It does not

Change 470: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: component values or device models, so its SPICE export is illustrative and uses placeholder values; simulation of real scans is not evaluated. The component-pair metric ignores terminal identity: a perfect score does not certify pin assignments, the exact net partition or the absence of shorts between pins of one component. The completion guard blocks new same-component merges but does not repair merges made

Change 471: replace | old pp. [8] | new pp. [10]

Before (including deletions): an independent, context-free model call (one per image), so no answer leaks between circuits,

After: the base graph. Table 11 lists the device groups: pin templates for transformers, optocouplers and other complex devices exist but are not validated,

Change 472: replace | old pp. [8] | new pp. [10, 11]

Before (including deletions): human, not

After: detector merges several original classes into one label. Data. All images come from one corpus. The human-verified nets were bootstrapped from a conservative endpoint-graph join and corrected by a single annotator, which may 10 VOLUME 11, 2023 TABLE 11. Connectivity evaluation, pin construction and export by device group. Device group Evaluated scope Pin construction Export R/C/L/D/Q, voltage sources and ICs Component-pair connectivity on both benchmarks Geometric templates; endpoint clustering for discrete and source types, generic layouts for ICs Placeholder values; ICs are skipped because no export model is implemented. Switches Not in

Change 473: replace | old pp. [8] | new pp. [11]

Before (including deletions): model, remains

After: evaluated electrical subset Two pins at

Change 474: replace | old pp. [8] | new pp. [11]

Before (including deletions): verifier of record. On

After: shorter OBB-edge midpoints; long-axis fallback A 0.001 Ω resistor if the two pins map to distinct nodes; no switch state is recovered. Transformers, thyristors, optocouplers and other complex devices Not evaluated Two-pin geometric guess for transformers; optocoupler template; generic fallback No export model; skipped. leave

Change 475: replace | old pp. [8] | new pp. [11]

Before (including deletions): clean synthetic control (15 authored circuits)

After: bias toward graph-based methods;

Change 476: replace | old pp. [8] | new pp. [11]

Before (including deletions): VLM scores F1 = 0.99, confirming it understands

After: held-out bench- mark, with labels from

Change 477: replace | old pp. [8] | new pp. [11]

Before (including deletions): task and symbol set. On all 31 real scans, evaluated against

After: dataset's own annotations, shows

Change 478: replace | old pp. [8] | new pp. [11]

Before (including deletions): human-verified net-level ground truth, it reaches micro-F1 = 0.923 (P = 0.97, R = 0.88; macro-F1 = 0.949). The VLM thus edges our deterministic join on aggregate (0.923 vs 0.890 micro-F1), but the gap

After: same ranking. That reference

Change 479: replace | old pp. [8] | new pp. [11]

Before (including deletions): statistically significant:

After: human-verified, mainly misses connections, and its quality filter and

Change 480: replace | old pp. [8] | new pp. [11]

Before (including deletions): paired bootstrap over

After: junction rule were fixed after inspecting the 17 overlapping images. The benchmark copies of

Change 481: replace | old pp. [8] | new pp. [11]

Before (including deletions): puts

After: are resized, partly re-oriented, and half stroke maps, which makes them easier than

Change 482: replace | old pp. [8] | new pp. [11]

Before (including deletions): dif- ference at +0.033 with

After: photographs. The Hough and CCL baselines keep their 31-image settings on the held-out photographs. Scale. The join tolerances derive from one median scale per image, and their clamps are tuned to native resolution. A component much smaller than the median can create

Change 483: replace | old pp. [8] | new pp. [11]

Before (including deletions): 95% CI of [-0.009, +0.078], which includes zero. Two further findings qualify

After: spuri- ous pair, as in

Change 484: replace | old pp. [8] | new pp. [11]

Before (including deletions): comparison. First, the VLM's

After: synthetic Wheatstone case. The extractor's pixel-valued parameters limit

Change 485: delete | old pp. [8] | new pp. []

Before (including deletions): degrades with circuit complexity: it is exact (F1 = 1.00)

After: (none)

Change 486: replace | old pp. [8] | new pp. [11]

Before (including deletions): 21

After: downscaled inputs. Rescaling tests changes in zoom, not in drawing style, and uses annotated components. Detector. No manifest

Change 487: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: detector's 85/15 random training split survives. The

Change 488: replace | old pp. [8] | new pp. [11]

Before (including deletions): circuits, including all the small 3–5-component cases, but on the largest boards it systematically omits connections, with recall falling to 0.63–0.68 on the three densest boards, exactly where a netlist error is most costly; its precision otherwise stays high (0.97 micro), so its failure mode is dropped nets rather than shorts. The macro-F1 (0.949) is correspondingly inflated by the many easy small circuits, which is why we report the pair-weighted micro-F1 as primary. Second, the cost

After: benchmark drawings come from drafters eligible for training,

Change 489: replace | old pp. [8] | new pp. [11]

Before (including deletions): form

After: about 26

Change 490: replace | old pp. [8] | new pp. [11]

Before (including deletions): the output differ from the geometric pipeline: each image consumes on the order of 105 tokens, the result is free-form text rather than a structured netlist, and nothing

After: them would be expected

Change 491: replace | old pp. [8] | new pp. [11]

Before (including deletions): model guarantees electrical validity (e.g. that a two-terminal component is not shorted). Our geometric join is statistically indistinguishable from

After: training split, so

Change 492: replace | old pp. [8] | new pp. [11]

Before (including deletions): VLM on these circuits while being deterministic, two to three orders of magnitude cheaper, structurally guaranteed against shorts by the self-loop guard, and simulatable by construction. We therefore treat the VLM as an upper reference

After: end-to-end numbers are likely optimistic

Change 493: replace | old pp. [8] | new pp. [11]

Before (including deletions): what a frontier model can do here, not as a connectivity module to ship: an explicit model reaches the same accuracy and also gives the determinism, low cost, and structural validity a deployed digitizer needs. V. DISCUSSION

After: detection.

Change 494: delete | old pp. [8] | new pp. []

Before (including deletions): endpoint-graph model resolves ambiguity that radius-based methods cannot: by representing both wire endpoints and component pins as nodes with typed edges, it distinguishes between a wire that passes near a component (endpoint–wire-body edge) and one that connects to it (endpoint–pin edge with directional preference). The component-first assignment further disambiguates by using the geometric constraint that wires must reach a component's pin, not just its bounding box. Degree-budget completion addresses the fundamental limitation of

After: (none)

Change 495: replace | old pp. [8] | new pp. [11]

Before (including deletions): noise: even a 97.6% accurate wire detector drops ~2% of wire endpoints, and those dropped endpoints are precisely the connections that matter most for simulation accuracy. The b-matching formulation bounds the damage from mis-detection (each floating pin gets at most one recovery edge), while the self-loop guard prevents the catastrophic short-circuit failures that plague unconstrained approaches. The human-verified net-level evaluation, while novel for hand-drawn circuits, currently covers 31 images; the 95% bootstrap confidence intervals on the per-strategy F1 are correspondingly wide (± 0.03 – 0.05) and overlap between adjacent strategies, so while the top ranking is consistent and corroborated by the independently authored synthetic suite, broader annotation is needed to separate the closely ranked

configurations. The net-level labels were bootstrapped by a conservative endpoint-graph join over perfect wires and then human-corrected; we mitigate the residual concern that this favors graph-based methods by validating on synthetic ground truth that is authored from scratch, where the same ranking holds. The evaluation draws from a single hand-drawn corpus (CGHD-1152); generalization across other drawing styles, capture conditions, and printed schematics remains to be validated, as does extension to circuits containing devices outside the simulatable set (switches, transformers, etc.), which we exclude from the net-level benchmark. The VLM connectivity comparison uses a single frontier model (Claude Opus 4.8) on the same 31 images; evaluating additional models and prompting strategies is future work. Component detection at 88.5% mean average precision (mAP@0.5) also leaves room for improvement, particularly on crossovers (70.7% recall). The component-detection model was trained on CGHD-1152 after excluding

After: excludes

Change 496: replace | old pp. [8] | new pp. [11]

Before (including deletions): diverged markedly

After: differs

Change 497: replace | old pp. [8] | new pp. [11]

Before (including deletions): other drafters; this exclusion reduces

After: rest of

Change 498: replace | old pp. [8] | new pp. [11]

Before (including deletions): training set but avoids domain-shift artifacts from an outlier style. Finally, the

After: corpus, and it is sensitive to orientation. Coverage. The human-verified benchmark contains 13 annotated crossovers, and we did not evaluate dense multi-track buses or multi-layer crossings. The VLM reference uses one model and one prompt. The

Change 499: replace | old pp. [8] | new pp. [11]

Before (including deletions): a first-pass approximation of detector noise rather than a

After: authored and not

Change 500: delete | old pp. [8] | new pp. []

Before (including deletions): fit; with the new real-image net-level metric available, calibrating it

After: (none)

Change 501: delete | old pp. [8] | new pp. []

Before (including deletions): the

After: (none)

Change 502: replace | old pp. [8] | new pp. [11]

Before (including deletions): failure distribution is

After: errors. Expansion. The derived reference makes

Change 503: replace | old pp. [8] | new pp. [11]

Before (including deletions): natural next step. VI.

After: larger human-verified benchmark affordable: its nets can seed the verification tool, which logs a timestamp at every save so that per-image effort can be measured. We plan to sample new images stratified by component count, drafter, crossing and bus structure and capture conditions, have two annotators correct each image independently with adjudication of disagreements, and extend the evaluation to printed schematics and to corpora other than CGHD. VII.

Change 504: replace | old pp. [8] | new pp. [11]

Before (including deletions): present

After: presented

Change 505: delete | old pp. [8] | new pp. []

Before (including deletions): complete

After: (none)

Change 506: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: structural netlist recovery from

Change 507: replace | old pp. [8] | new pp. [11]

Before (including deletions): circuit digitization,

After: schematics,

Change 508: replace | old pp. [8] | new pp. [11]

Before (including deletions): occlusion-first

After: occlusion- first

Change 509: replace | old pp. [8] | new pp. [11]

Before (including deletions): extrac- tor that yields an endpoint representation and

After: extractor,

Change 510: replace | old pp. [8] | new pp. [11]

Before (including deletions): two-stage endpoint-graph join, together with, to our knowledge, the first human-verified net-level connectivity benchmark for hand- drawn circuits. The

After: typed

Change 511: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: and degree- budget completion. With annotated components, the join reaches component-pair micro-F1 0.890 on 31 human- verified images and 0.711 on 164 held-out photographs

Change 512: replace | old pp. [8] | new pp. [11]

Before (including deletions): degree-budget completion is most robust under synthetic error injection (F1 = 0.95

After: independently derived reference nets, and it beats every deter- ministic baseline on both with significant paired margins. Its accuracy holds from 0.5× to 3× image scale when wires are given, and it reaches 0.627 end to end with a trained detector. A VLM given the same components scores higher per image,

Change 513: replace | old pp. [8] | new pp. [11]

Before (including deletions): highest severity vs. 0.36 for

After: cost of a proprietary call per image. Better extraction on difficult photographs and better component detection are

Change 514: replace | old pp. [8] | new pp. [11]

Before (including deletions): radius base- line), and, paired with a high-precision scale-relative base, attains connectivity micro-F1 = 0.890 on real images against human-verified ground truth, beating the prior completion default (0.83) and every deterministic baseline, including the connected-component net-tracing recipe used by recent systems (0.62) and a Hough-plus-proximity line tracer (0.81), all on identical detected wires. Running the join on perfect wires does not raise micro-F1 at all, showing the join, not wire detection, is the bottleneck. A first-party experiment with a frontier VLM reaches 0.923 on the same images, a difference from our join that is not statistically significant (paired 95% CI [-0.009, +0.078]), but at two

After: most direct routes

Change 515: delete | old pp. [8] | new pp. []

Before (including deletions): three orders of magnitude 8 VOLUME 11, 2023

After: (none)

Change 516: replace | old pp. [9] | new pp. [11]

Before (including deletions): cost and without simulatable, structurally valid output, supporting the explicit geometric design. Future work includes scaling the human-verified benchmark, calibrating the synthetic error model to the measured detector distribution, and extending coverage to devices outside the simulatable set.

After: end-to-end accuracy.

Change 517: replace | old pp. [9] | new pp. [11]

Before (including deletions): IV-F,

After: V-I,

Change 518: replace | old pp. [9] | new pp. [11]

Before (including deletions): evaluation scripts,

After: the

Change 519: replace | old pp. [9] | new pp. [11]

Before (including deletions): genera- tor,

After: generator,

Change 520: replace | old pp. [9] | new pp. [11]

Before (including deletions): (ground_truth/real_nets_verified.json),

After: (ground_truth/real_net

Change 521: replace | old pp. [9] | new pp. [11]

Before (including deletions): join- strategy

After: CGHD-derived reference nets of the held-out bench- mark (ground_truth/cghd_ref/), the join-strategy

Change 522: delete | old pp. [9] | new pp. []

Before (including deletions): connected-component

After: (none)

Change 523: delete | old pp. [9] | new pp. []

Before (including deletions): and

After: (none)

Change 524: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: and the evaluation scripts for the paired tests, strata, rescaling, held-out and end-to-end experiments (wire_detection/benchmark/revision2/)

Change 525: replace | old pp. [9] | new pp. [11]

Before (including deletions): avail- able

After: available

Change 526: replace | old pp. [9] | new pp. [11]

Before (including deletions): <https://github.com/boscochanam/circuit-digitization>.

After: <https://github.com/boscochanam/circuit-digitization>, together with the image-to-drafter mapping used in Table 10.

Change 527: replace | old pp. [9] | new pp. [11]

Before (including deletions): component-detection

After: reference nets are derived from the CGHD annotations [1]; no CGHD images are redistributed. The component-detection

Change 528: replace | old pp. [9] | new pp. [11]

Before (including deletions): <https://huggingface.co/boscochanam/circuit-component-detector>,

After: <https://huggingface.co/boscochanam/circuit-component-detector>,

Change 529: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: VOLUME 11, 2023 11

Change 530: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: S. Gao, F. Shi, X. Xu, J. Chen, H. H.-C. Lu, et al., “A memristor-based neural network circuit with classical conditioning and fear generalization,” IEEE Trans. Consumer Electron., 2026, doi: 10.1109/TCE.2026.3663359. [7] S. Gao, Y. Song, Y. Cao, H. H.-C. Lu, L. Qin, Y. Zhang, and J. Mou, “Biologically plausible memristive decision-making circuit for adaptive control in industrial autonomous navigation,” IEEE Trans. Ind. Informat., vol. 22, pp. 4801–4812, 2026, doi: 10.1109/TII.2026.3668794. [8]

Change 531: replace | old pp. [9] | new pp. [12]

Before (including deletions): [7]

After: [9]

Change 532: replace | old pp. [9] | new pp. [12]

Before (including deletions): [8]

After: [10]

Change 533: replace | old pp. [9] | new pp. [12]

Before (including deletions): [9]

After: [11]

Change 534: replace | old pp. [9] | new pp. [12]

Before (including deletions): [10]

After: [12]

Change 535: replace | old pp. [9] | new pp. [12]

Before (including deletions): [11]

After: [13]

Change 536: replace | old pp. [9] | new pp. [12]

Before (including deletions): [12]

After: [14]

Change 537: replace | old pp. [9] | new pp. [12]

Before (including deletions): [13]

After: [15]

Change 538: replace | old pp. [9] | new pp. [12]

Before (including deletions): [14]

After: [16]

Change 539: replace | old pp. [9] | new pp. [12]

Before (including deletions): [15]

After: [17]

Change 540: replace | old pp. [9] | new pp. [12]

Before (including deletions): [16]

After: [18]

Change 541: replace | old pp. [9] | new pp. [12]

Before (including deletions): [17]

After: [19]

Change 542: replace | old pp. [9] | new pp. [12]

Before (including deletions): [18]

After: [20]

Change 543: replace | old pp. [9] | new pp. [12]

Before (including deletions): [19]

After: [21]

Change 544: replace | old pp. [9] | new pp. [12]

Before (including deletions): [20]

After: [22]

Change 545: replace | old pp. [9] | new pp. [12]

Before (including deletions): [21]

After: [23]

Change 546: replace | old pp. [9] | new pp. [12]

Before (including deletions): [22]

After: [24]

Change 547: replace | old pp. [9] | new pp. [12]

Before (including deletions): [23]

After: [25]

Change 548: replace | old pp. [9] | new pp. [12]

Before (including deletions): [24]

After: [26]

Change 549: replace | old pp. [9] | new pp. [12]

Before (including deletions): [25]

After: [27]

Change 550: replace | old pp. [9] | new pp. [12]

Before (including deletions): [26]

After: [28]

Change 551: delete | old pp. [9] | new pp. []

Before (including deletions): VOLUME 11, 2023 9

After: (none)

Change 552: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: 12 VOLUME 11, 2023

Change 553: delete | old pp. [10] | new pp. []

Before (including deletions): 10

After: (none)

Change 554: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: 13